Review-Aware Recommender Systems (RARSs): Recent Advances, Experimental Comparative Analysis, Discussions, and New Directions

GUILHERME BITTENCOURT, Universidade Federal de São João del-Rei, Brazil
NAAN VASCONCELOS, Universidade Federal de São João del-Rei, Brazil
YAN ANDRADE, Universidade Federal de São João del-Rei, Brazil
NÍCOLLAS SILVA, Universidade Federal de Minas Gerais, Brazil
WASHINGTON CUNHA, Universidade Federal de Minas Gerais, Brazil
DIEGO DIAS, Universidade Federal do Espírito Santo, Brazil
MARCOS ANDRÉ GONÇALVES, Universidade Federal de Minas Gerais, United States
LEONARDO ROCHA, Universidade Federal de São João del-Rei, Brazil

Recommender systems (RSs) have emerged as an effective strategy for dealing with information overload. Their importance is undeniable as they are widely adopted in various web applications, presenting the potential to solve various problems associated with the abundance of choices. In recent years, the literature has witnessed the proposal of numerous recommendation techniques, constantly seeking to improve the effectiveness of these methods. Although many approaches have focused on techniques that employ quantitative (numerical) evaluations, it is essential to recognize the significant value of the *user feedback*, usually expressed as (textual) comments (a.k.a., reviews) in order to understand their preferences. In this context, various algorithms have been developed to effectively take advantage of comments as a valuable source of information for Review-Aware RSs (RARSs), capable of generating recommendations in line with each user's profile using their respective comments. This article aims to review the primary research efforts on RARSs comprehensively. We present a taxonomy of recommendation models in this domain, accompanied by a detailed summary of the main advances, the main datasets, and the used metrics. In addition, we conduct a comprehensive experimental comparison among state-of-the-art proposals, highlighting the main directions and new perspectives for future developments. Code and datasets of our open benchmark are available at https://github.com/guibitten03/iRev for fostering replicability and new advances.

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Authors' Contact Information: Guilherme Bittencourt, Universidade Federal de São João del-Rei, Sao Joao del-Rei, MG, Brazil; e-mail: bittenkourt.gmf@aluno.ufsj.edu.br; Naan Vasconcelos, Universidade Federal de São João del-Rei, São João del Rei, MG, Brazil; e-mail: naan.vasconcelos@aluno.ufsj.edu.br; Yan Andrade, Universidade Federal de São João del-Rei, Belo Horizonte, Brazil; e-mail: yan.ribeiro@dcc.ufmg.br; Nícollas Silva, Universidade Federal de Minas Gerais, Belo Horizonte, MG, Brazil; e-mail: ncsilvaa@dcc.ufmg.br; Washington Cunha, Universidade Federal de Minas Gerais, Belo Horizonte, Brazil; e-mail: washingtoncunha@dcc.ufmg.br; Diego Dias, Universidade Federal do Espírito Santo, Vitoria, Brazil; e-mail: diego.dias@ufes.br; Marcos André Gonçalves, Universidade Federal de Minas Gerais, Belo Horizonte, United States; e-mail: mgoncalv@dcc.ufmg.br; Leonardo Rocha, Universidade Federal de São João del-Rei, Sao Joao del-Rei, MG, Brazil; e-mail: lcrocha@ufsj.edu.br.

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CCS Concepts: • **Information systems** → **Recommender systems;**

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1 Introduction

In the face of the vast expanse of online information, **Recommender Systems** (**RSs**) have emerged as crucial tools in simplifying decision-making processes and enhancing user experiences. Whether by facilitating searches for music tracks [18], movies [32], or even aiding in identifying social connections [22], RSs have become omnipresent components of everyday life. Fundamentally, RSs endeavor to discern users' interests and preferences through explicit interactions or implicit information gleaned from their behaviors, captured while using the system [69, 123]. This personalization process enables content and recommendations to be finely tailored to meet users' needs. However, the effectiveness of personalization hinges upon the system's capacity to comprehend and accurately represent user preferences [122, 123, 155].

Researchers have directed their attention toward explicit user evaluations of items and recommendations, commonly known as reviews, more commonly expressed in textual form. Models that employ such information, here referred to as **Review-Aware Recommender Systems** (**RARSs**) [2, 25, 136], aim to enhance the quality of personalized recommendations by leveraging user ratings and opinions as valuable sources of information. The fundamental premise underlying these models is that users' (textual) opinions tend to reflect more reliably their genuine perceptions of an item's relevance [106]. While RARSs are not a novel concept and have been the subject of investigation for some time [62, 139], their resurgence has been fueled by notable advancements in neural network architectures, especially **deep neural network** (**DNN**) learning applied to **natural language processing** (**NLP**) tasks, including automatic classification [34, 36–38, 117] and **topic modeling** (**TM**) [143].

Recently, an increasing number of RSs have been incorporating NLP methodologies into their inner workings, exploring diverse architectural paradigms. These encompass Attention-Based Networks [86], **Temporal Convolutional Networks** (**TCN**) [13], and graph-based approaches [146]. Furthermore, NLP techniques such as aspect extraction [26] and TM [138, 142], along with target evaluation [163], Sentiment-Based analysis [30], and, more recently, contrastive learning [131], have been increasingly integrated into RSs. This surge in research activity is evident from the number of recent publications in top-tier conferences on RSs and computing worldwide. The noteworthy success of RARSs, spanning from academia to industry, necessitates a comprehensive review and synthesis to facilitate a deeper understanding of their potential and limitations and the contextual nuances in which these models are deployed. There is a pressing need to identify common principles and elucidate their interplay, fostering more robust and effective recommendation strategies.

1.1 Research Objectives

The primary objective of this article is to conduct a comprehensive survey of the most recent developments in RARSs, a field that has undergone significant evolution in recent years but has not been comprehensively surveyed yet. To this end, this article presents a systematic mapping of RARSs studies with the following goals: (i) to provide an updated overview of the principal research endeavors conducted in this domain for the benefit of future investigations; (ii) to delineate the primary limitations, characteristics, and community-driven guidelines that inform our collective efforts. Notably, the two last studies with similar objectives were conducted in 2019 and 2020 [2, 136], focusing specifically on integrating sentiment analysis into RSs utilizing user comments [2]. Both studies also do not consider new and recent neural architectures [136].

Only in 2015 [25], way prior to the recent DNN revolution applied to NLP, it is possible to find a systematic mapping of the literature that also addressed the topic explored in this article. Specifically, in this current study, we have curated and categorized a corpus of 124 papers from 2014 to 2024, thereby establishing a comprehensive 10-year search window. The following research questions guide our analysis of these papers:

— RQ1: How do recommendation algorithms employ NLP techniques to infer user preferences from textual comments?
— RQ2: What are the predominant datasets and evaluation metrics utilized in studies about this domain?
— RQ3: How are experimental evaluations conducted in the scrutinized studies, considering the current state-of-the-art configurations and model parameters?

Building upon the insights gathered by addressing these questions, our secondary objective is to propose an unclosed and extensible RARSs benchmark to be considered in evaluating future proposals and conduct a comprehensive comparative analysis of the principal RARSs. We aim to exhaustively examine the quality metrics and datasets utilized in the evaluation methodologies proposed across the literature, delineating the strengths and weaknesses of the assessed strategies, thereby elucidating opportunities for enhancement and innovation in subsequent proposals.

1.2 Overview of the Main Findings

Our initial contribution is the proposal of a novel taxonomy tailored to the pertinent domain. This taxonomy is structured around four overarching perspectives: (i) Data Modeling, (ii) Information Extraction; (iii) Model Objective; and (iv) Learning Architecture.

The **first** perspective pertains to the strategy for modeling textual information, wherein we delineate three distinct classes: (1) *Document Modeling*, (2) *Sentence Modeling*, (3) *Rating Aggregation*. The **second** perspective revolves around extracting information during the construction of RSs. Within this benchmark, we propose categorizing RSs into two distinct classes: (1) *Sentiment Extraction* and (2) *Aspect Extraction*. The **third** perspective concerns the purpose of the recommendation model: (1) *predictive*, which makes recommendations; (2) *explanatory*, aimed at explaining the recommendations; and (3) *hybrid*, which do both. The **fourth** and final perspective concerns the learning process itself. Most articles use **artificial neural network** (**ANN**) architectures to process the texts and/or make the recommendations themselves. For this perspective, we propose classifying the works into two distinct classes: (1) *non-neural models*, which use **matrix factorization** (**MF**), clustering, amongst other techniques to make recommendations [110, 138]; and (2) *neural models*.

Concerning RQ2 and RQ3, the most used data collections are: (1) *Amazon's product dataset* and *Yelp's points of interest dataset*, as they provide user/item interactions and comments arising from the interactions. Regarding evaluation metrics, error metrics are the most widely used forms of evaluation, followed by ranking evaluation metrics. Despite the consensus in the RS community that accuracy-based metrics alone are not enough to evaluate RSs, the vast majority of studies prioritize accuracy over other quality dimensions, such as serendipity and diversity.

Another issue concerns the algorithms that are considered state-of-the-art and their configurations. There is no consensus on the baselines to be considered; each article uses a different set, and parameter settings are rarely reported. Less than 50% of the papers analyzed provide source code for their proposals, and less than 30% provide the proposed algorithms' parameter settings.

In order to fill this gap in the literature, we propose and implement an experimental RARSs benchmark composed of six different datasets from Amazon and Yelp applications and eight metrics related to different quality dimensions: (1) error-based (i.e., **Mean Square Error** (**MSE**), **Root Mean Square Error** (**RMSE**), **Mean Absolute Error** (**MAE**)); (2) ranking-based (i.e., Recall, nDCG); and novelty/diversity (i.e., novelty and categorical diversity). We adopt a methodology that assigns a score to each algorithm proposed in the literature based on the number of citations and the baselines, data collections, and metrics used in experimental analysis and recency. Based on the application of this methodology, we selected and implemented the 16 proposed algorithms with the highest scores. All these artifacts have been made publicly available at https://github.com/guibitten03/iRev, which can be used as a benchmark for future work. The benchmark is also open to new contributions (i.e., new algorithms, collections, and metrics).

We conduct a comprehensive comparative evaluation utilizing all 16 algorithms. We observed that no algorithm excelled in all scenarios—the most effective alternatives in one type of metric obtained poor or average results in the others. At the data collection level, none achieved the best results for more than three metrics in more than three data collections. The results highlight the need to evaluate the systems in different and complementary scenarios to ensure sound scientific and practical advances in the RARS scenario. Our benchmarking is a valuable, concrete contribution toward this goal.

1.3 Main Contributions

We examine the literature on advances in RSs based on user reviews – RSARs – providing an overview so that new researchers can understand the landscape of the field. This study lays the foundations for driving innovation in RSARs, exploring the breadth of this research domain. The research aims to provide fundamental guidelines aimed at researchers, practitioners, and educators interested in RARSs. In summary, the main contributions of this study are:

- We conducted a comprehensive review of recommendation models anchored in user comment analysis, including issues related to key datasets and evaluation;
- We propose a new classification taxonomy to position and organize existing (and future) literature;
- We carry out a comprehensive experimental comparative study considered the most representative RARSs models;
- We create an open and extensible RARSs benchmark to be considered in the evaluation of future proposals, with all the implementations of the compared models; and
- We discuss challenges, outstanding issues and identify new trends and future directions in RARSs research field.

2 The Systematic Mapping Process

We discuss the process of collecting, filtering, and selecting articles for categorization and experimental evaluation inspired by the methodology used by [39, 132]. This protocol for carrying out a systematic mapping consists of three phases: (1) defining the search words based on the research questions and collecting articles from the main literature sources; (2) selecting the relevant articles according to evaluation criteria; and (3) extracting the main information contained in the articles considered as relevant. The complete process can be seen in Figure 1.

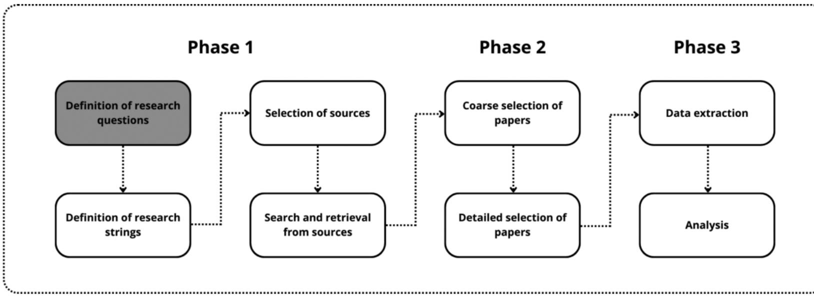

Fig. 1. Process of selecting studies and analyzing the literature.

2.1 Phase 1: Research Questions, Search Words, and Digital Sources

We have formulated three primary research questions that serve as the focal points of this **Systematic Literature Review** (**SLR**):

— **RQ1:** How do recommendation algorithms employ NLP techniques to infer user preferences from textual comments?
— **RQ2:** What are the predominant datasets and evaluation metrics utilized in the RARSs domain?
— **RQ3:** How are experimental evaluations conducted in the scrutinized studies, considering the current state-of-the-art, configurations and model parameters?

We employed the Google Scholar search engine to conduct three queries, initiating our initial article collection. We selected Google Scholar for two main reasons: (i) its extensive coverage, which includes digital libraries from prestigious publishers such as ACM, IEEE, Springer, and Elsevier, and (ii) the provision of information regarding the number of citations for each research article, an important criterion for our method selection process. The queries (search terms) utilized for the collection are enumerated below:

— **SS-Q1:** (“review based” OR “review aware” OR “review modeling”) AND (“recommender systems" OR “recommendation systems" OR “recommender system”) AND (“text” OR “textual" OR “review”)
— **SS-Q2:** (“review based” OR “review aware” OR “review modeling”) AND (“recommender systems“ OR “recommendation systems“ OR “recommender system“) AND (“evaluation” OR “measure" OR “metrics”)
— **SS-Q3:** (“review based” OR “review aware” OR “review modeling”) AND (“recommender systems" OR “recommendation systems" OR “recommender system”) AND (“source code” OR “reproducibility” OR “empirical” OR “experimental”)

2.2 Phase 2: Selecting Relevant Articles

Articles were collected from Google Scholar using a date filter from 2014 to 2024, thus ensuring a scope covering 10 years of publications. Upon completion of the collection process, a total of 1,196 articles were accumulated. Initially, a preliminary screening was conducted to eliminate duplicates, resulting in a remaining count of 832 articles. A second filtering criterion was applied concerning conference venues, wherein only articles presented at the top 100 conferences and journals—defined as those with the highest impact factor and relevance in Computer Science, as per the rankings published by Research.com—were considered. These venues include RecSys, WWW, WSDM, KDD, SIGIR, Neurocomputing, among others. Following the venue-based filtering, 681 articles remained.

Based on this refined set of articles, an additional, more specific filtering criterion was applied to facilitate a segmented analysis. More specifically, we employ an advanced filtering approach incorporating both inclusion and exclusion criteria to refine the 681 articles and select those pertinent to our research objectives. Each article underwent a meticulous manual analysis, wherein it was classified as either relevant or non-relevant, according to the established criteria:

Inclusion Criteria

— The primary approach for recommendation entails leveraging user comments, with numerical evaluations serving as supplementary support if used at all.
— Articles introduced relevant advancements and innovations within the field, extending beyond merely optimizing pre-existing algorithms.
— Comparative experimental evaluations are conducted between previous approaches utilizing user comments and the methods proposed in the respective articles. The proposed algorithms exhibit improvements over the baselines.

Exclusion Criteria

— Apart from comments, they incorporate other sources of information such as images, audio, or video to predict user preferences—i.e., they focus on multimodal algorithms.
— They encompass surveys, case studies, and systematic or experimental reviews of algorithms within the domain, devoid of any innovative propositions or novel strategies in the realm of recommendation.
— They utilize comments solely to justify recommendations, focusing on explainability. They do not directly integrate comments into the inner workings of the recommendation strategy.

After applying the previously mentioned criteria, 124 articles[1] were identified as most relevant for the systematic mapping. These articles form the core of our analyses.

2.3 Phase 3: Extracting Information from Articles

Following the completion of the article collection and processing phase, we conducted a thorough examination of all 124 remaining articles. That involved identifying the principal characteristics of the proposed solutions, elucidating their key innovations and contributions to the literature, and analyzing the evaluation methodologies employed. The subsequent sections delineate the results of this analysis, aiming to address the three research questions posed at the outset of this section.

3 A New Taxonomy for NLP in Recommendation Systems

We start by addressing the first research question: **How do recommendation algorithms employ NLP techniques to infer user preferences from textual comments?** To achieve this objective, we developed a novel taxonomy to analyze and categorize each of the 124 articles identified in our SLR. This taxonomy begins by categorizing the articles into four primary perspectives:

(1) **Data Modeling:** discusses how textual data and user ratings are pre-processed and modeled.
(2) **Information Extraction:** discusses how information regarding user preferences and item characteristics are extracted from textual data.
(3) **Model Purpose:** discusses the purpose of using the information extracted from comments either in the recommendation itself or in the explanation of the recommendation.
(4) **Learning Architecture:** discusses the **machine learning** (**ML**) techniques for achieving the desired goal with the extracted information.

These perspectives facilitate a comprehensive analysis of the complete process typically employed by all articles investigating RARSs. As depicted in Figure 2, we scrutinized every aspect,

[1] https://docs.google.com/spreadsheets/d/1WAp7J67QqoRB2wdJCdTTi3SGcYmafY6krVI2cw9sydw/edit?gid=0#gid=0

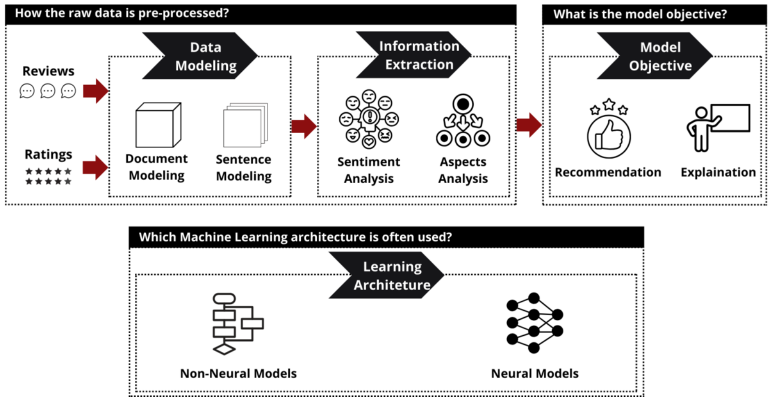

Fig. 2. Graphical representation of the perspectives proposed for the first level of the taxonomy.

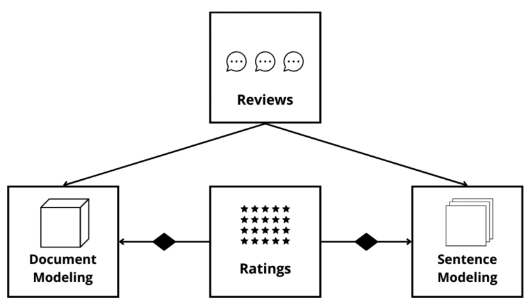

Fig. 3. Graphical representation of the modeling techniques.

ranging from data modeling to the architecture of the ML model utilized to accomplish our goals. Our discussions and observations provide valuable insights to researchers interested in all stages of the process. The taxonomy can be readily expanded and adjusted to incorporate new categories.

3.1 Data Modeling

Despite the progress made with Transformers and **Large Language Models** (**LLMs**), recent studies show that text modeling is still one of the main factors responsible for the results of NLP models [44]. In this context, our first perspective of analysis investigates how the selected articles (pre-)process and model textual data and user evaluations. Our analyses identified three distinct categories represented in Figure 3, consisting of: (1) **Document Modeling:** works that concatenate all user comments, including on items, into a single document; (2) **Sentence Modeling:** works that propose mechanisms for recognizing key words in sentences and, subsequently, key sentences, to compose the representation of users and items; and (3) **Rating Aggregation:** works that seek to associate the numerical evaluations of users/items with their respective comments, similar to traditional RSs that rely on collaborative filtering. We detail each of them below.

The first category under analysis, **Document modeling**, involves creating a single document to represent each user/item within the recommendation scenario. Typically, this approach entails concatenating all the comments from each user or employing embedding pooling of the representations of all user comments. Similarly, the process is repeated for items, often incorporating additional textual information such as item names or descriptions. One of the primary advantages of document modeling is its simplicity in extracting characteristics, as all user comments can be processed simultaneously. That not only facilitates information modeling but also mitigates the computational demands. On the other hand, a notable drawback is that it fails to consider individual comments, resulting in a loss of information regarding the unique characteristics of each comment. Prominent examples of this approach include the CARL [159] and ANR [30] models, widely utilized as baselines, which employ this strategy for text data modeling. DeepCoNN [189], introduced in 2017 and the most cited work in the field, adopts a similar approach by processing the documents representing each user and item using **convolutional neural networks (CNNs)**.

Conversely, certain studies [17, 21] critique the practice of representing data as a single document due to the potential loss of information and the risk of information leakage during algorithm training. These studies advocate for a separate treatment of comments, emphasizing the strong correlation between comments and user-item interaction. We categorize this group of articles as **Sentence Modeling**, which propose mechanisms for identifying keywords within sentences and, subsequently, key sentences to represent users and items. Notable examples within this category include NARRE [21], DAML [94], and TransNet [17], widely used as baselines.

The NARRE algorithm [21] employs attention mechanisms to identify the primary words within each comment and represents users/items based on the set of most significant words. It then combines the user's representation with that of the item and utilizes neural factorization machines to perform successive dimensionality reductions, ultimately estimating the user's preference for the item. DAML [94] also employs attention mechanisms to extract the most important words from comments. Following the creation of representations, DAML utilizes an affinity matrix concept to assess the similarity between the user's most significant words and those of the items. Subsequently, fully connected layers are utilized to reduce the dimensionality of the matrix to a single value. TransNet [17], on the other hand, applies convolutional layers and pooling techniques to reduce the dimensionality of sentences, concatenating all user/item sentences to obtain their representations. These representations are then concatenated, and fully connected layers are employed to reduce the dimensionality further into a single value.

Additionally, there is a discussion in the literature regarding integrating information from user comments with their numerical ratings. We classify this category as **Rating Aggregation**, as these approaches aim to link the explicit feedback provided by users with user/item representations. These algorithms resemble traditional collaborative filtering-based RSs. ConvMF [78] stands as a notable example within this category. A pioneer in utilizing convolutional layers within neural networks in the RARS domain, ConvMF employs MF to derive a latent user representation from their matrix of numerical ratings. Subsequently, it computes the user's affinity with item representations generated from the provided comments.

Another exemplary algorithm is TARMF [104], which leverages representations of users and items learned from numerical evaluations. It computes an approximation between the representations generated by the documents to adjust the algorithm's parameters. More recent RARSs, such as CARM [86], calculate a confidence matrix and regress a confidence degree function from the numerical user and item rating distribution. This refinement process aims to balance the representations obtained from user-item interaction comments.

We categorize all 124 selected papers into the three described categories. Table 1 presents the classification results. It is important to note that the classification is not entirely disjoint, meaning that some articles may be related to more than one category. For instance, Zeng et al. [179] is classified as **document modeling** and **sentence modeling** because it concatenates all user/item comments into a document and employs convolutional layers to reduce dimensionality for creating its representation. Subsequently, it uses the same method to process the respective comment of the user-item interaction for which the rating will be predicted. Finally, it computes the error

Table 1. Organization of RARSs According to the Information Modeling Strategy

Modeling Method	Papers
Document Modeling (47%)	[5, 7, 9, 14, 19, 20, 26, 29, 30, 41, 45, 46, 51, 54, 54, 55, 58, 58, 67, 67, 68, 71, 76, 78, 85, 86, 88, 92, 93, 96, 104, 109, 112, 118, 128, 135, 137, 140, 147, 150, 151, 153, 154, 158, 159, 161, 164–166, 168–170, 172–174, 176, 179, 180, 185, 189]
Sentece Modeling (52%)	[3, 4, 12, 13, 15, 17, 21, 24, 27, 28, 31, 33, 40, 49, 50, 53, 56, 57, 59–61, 63, 70, 76, 77, 81–83, 87, 90, 94–98, 100–103, 107, 113, 115, 120, 121, 126, 127, 129–131, 134, 138, 146, 148, 149, 157, 160, 163, 167, 171, 175, 179, 182–184, 186]
Rating Aggregation (26%)	[4, 9, 15, 19, 26, 28, 31, 40, 51, 55, 71, 82, 86, 88, 92, 97, 100, 102, 104, 112, 129, 131, 137, 153, 161, 165, 167, 174, 180, 183, 186]

between the representations obtained by processing the document and the comment referring to the interaction, propagating errors in the algorithm, and adjusting parameters.

Similarly, Lu et al. [104], classified as **document modeling** and **rating aggregation**, also utilizes convolutional layers to reduce the dimensionality of the user/item document to acquire a representation. It then employs a collaborative filtering strategy to generate latent representations of users and items from the decomposition of the rating matrix. Finally, it calculates the difference between the representation of users/items obtained from the document and the representation obtained by factorizing the rating matrix and adjusting the system parameters based on the obtained differences.

3.2 Information Extraction

The second perspective of analysis discusses how to extract information on user preferences and item characteristics from textual data. This categorization segments the articles into how the information, once modeled, is extracted for building RARSs. Additionally, RARSs must present mechanisms to utilize this information in learning user preferences and, consequently, in the recommendation process. We propose categorizing RARSs into two distinct categories to achieve this goal, as shown in Figure 4: (1) **Sentiment Extraction:** This category encompasses articles that utilize sentiment analysis techniques to extract information from textual comments; (2) **Aspect Extraction:** includes articles proposing mechanisms for extracting latent and implicit aspects from textual comment. We detail each of them below.

3.2.1 Sentiment Extraction. This category comprises mostly articles that treat reviews as complete interactions initiated by users with specific items. These studies consider comments as reflections of user preferences, which are shaped by the feelings and emotions experienced by the user toward each item. Among the prevalent sentiment analysis methodologies, two main approaches for extracting preferences from comments stand out: **(A) Polarization** and **(B) Emotion**.

Polarization categorizes comments as positive, negative, or neutral, similar to traditional sentiment analysis in the NLP domain. This categorization, also termed as *helpfulness*, determines the significance of a comment when extracting user preferences. Consequently, this technique can also discern users who contribute relevant comments from those who provide irrelevant ones. For instance, Ashok et al. [5] employ sentiment analysis on user comments to discern sentimental associations with the item under evaluation, including potentially other users in case of a social interaction scenario. Similarly, Lei et al. [83] propose a system to forecast numerical ratings based on the sentiments implicit in comments within a social interaction dataset.

On the other hand, **Emotion**-based approaches aim to extract the specific emotion evoked in the user by the item, such as satisfaction, happiness, or frustration. These algorithms typically use

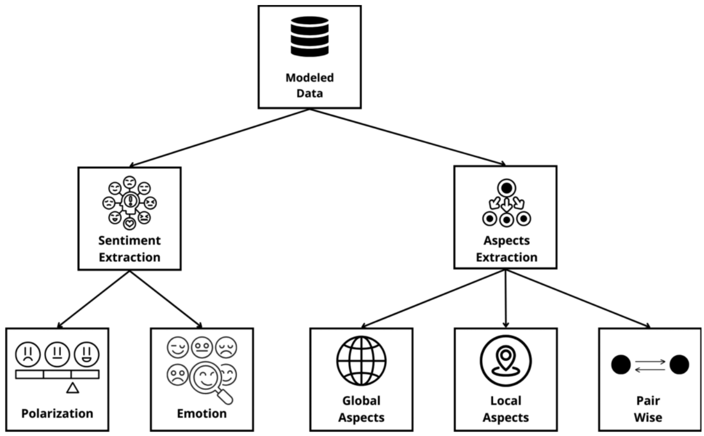

Fig. 4. Graphical representation of the information extraction techniques observed.

weighted predictions of numerical evaluations, assuming that sentiment analysis can influence the weight of the provided numerical rating in the interaction. They are often employed to gauge the degree of user affinity toward an item based on the emotions elicited by it. Unlike polarization, emotion-based approaches do not focus on classifying comments as useful or not; instead, they aim to extract the emotions experienced by the user when evaluating the item. For instance, Dong et al. [49] propose emotion extraction using word-level and sentence-level attention mechanisms and incorporate the Gumbel Softmax activation function to compute probabilities at the emotion level.

3.2.2 Aspect Extraction. For the second category of work, we identified proposals aimed at extracting latent characteristics that encapsulate the essence of comments within fewer dimensions. These RARSs employ intricate functions to deduce, from the comments, the aspects most pertinent to user preferences. Likewise, these models utilize this methodology to encapsulate the primary aspects highlighted by a particular item, thereby uncovering implicit characteristics of items within the dataset. These studies primarily diverge in the NLP techniques employed.

We have defined a subcategory of works that utilize TM to extract these aspects in a **(A) Global** manner. The fundamental premise of these algorithms is the assumption that within a given context, such as movie or book reviews, all comments associated with the products will converge on overarching and universal aspects about the primary topics discussed by users. Generally, users are represented by the topics (aspects) most closely related to the comments they provided, just as items are represented by the aspects most similar to the comments they received. Typically, these algorithms employ non-neural architectures. For instance, HFT [110] endeavors to estimate latent factors and topics from user comments using classical TM techniques. Conversely, A3NCF [28] integrates LDA-based TM [10] with attention layers to glean insights from comments.

A second group of papers propose strategies to extract aspects in a **(B) Local** manner. These algorithms operate under the premise that extracting global topics from the entire dataset induces bias. In other words, aggregating all comments and extracting topics disregards individual user preferences. Therefore, rather than analyzing all comments together, these algorithms advocate for separating comments by a user to ensure that individual preferences are respected. Primarily exploiting neural networks, these systems utilize convolutions and attention mechanisms to extract key features representing a user's preferences solely from their provided comments.

Scaling issues arise due to the fact that a user's preferences may not align with those of other users. TM does not require complex algorithms when aggregating comments from the entire dataset for

Table 2. Organization of RARSs According to the Information Analysis Methodology

Review Analysis Method	Review Element	Papers
Sentiment Analysis	Polarization (41%)	[3, 5, 7, 9, 15, 19, 20, 23, 26, 27, 29, 30, 33, 42, 43, 49, 51, 54, 59, 63, 67, 70, 71, 76, 77, 82, 83, 85, 88–90, 92, 103, 109, 112, 126, 129, 134, 135, 138, 147–150, 154, 157, 159, 168, 169, 175, 180, 183, 184]
	Emotion (10%)	[3, 19, 23, 49, 59, 63, 67, 71, 109, 157, 175, 183, 184]
Aspects Analysis	Global (21%)	[3, 7, 14, 15, 19, 28, 29, 42, 45, 68, 70, 71, 82, 83, 90, 92, 93, 104, 118, 135, 137, 138, 164, 167, 183–185]
	Local (54%)	[3–5, 9, 14, 15, 21, 23, 24, 26, 28–30, 33, 41–43, 49, 53, 55–57, 59, 63, 68, 77, 78, 81, 85–90, 94, 95, 100, 101, 103, 104, 109, 112, 113, 115, 120, 121, 128–130, 135, 137, 140, 153, 154, 157, 159, 160, 165, 171, 172, 174, 182, 183]
	Pair Wise (26%)	[4, 7, 14, 17, 24, 31, 49, 50, 53, 56, 59, 60, 67, 70, 77, 95, 97, 100, 102, 109, 115, 121, 131, 137, 153, 157, 158, 160, 163, 165, 170, 171, 175, 182, 183, 186]

aspect extraction. However, more sophisticated algorithms are necessary to extract user preferences from their respective comment sets and align them in the same space. Consequently, these systems employ neural projection mechanisms to align representations into a common latent space. For instance, Chin et al. [30] employ attention mechanisms for aspect extraction from comments, while Cheng [29] proposes ALFM, which extracts latent aspects from factorized rating matrices alongside aspects derived from user comments.

However, some authors argue that previous approaches overlook still another side of interactions. In the methods discussed earlier, user representations are solely based on comments and do not incorporate information about the item to which the comment pertains. Hence, **(C) Pairwise** systems contend that it is essential to consider the provided comment along with the associated item information to create a comprehensive user representation. Given the complexity of such representations, algorithms typically employ graph neural networks, enabling users to be represented by their comments (represented as edges) and the rated items (represented as nodes). Consequently, instead of using sets of comments for each user, these systems process each comment separately, as each review signifies the interaction between the user and a specific item. They understand that the aggregation of all comments provided by a user may not accurately represent their overall preferences; instead, each comment reflects the user's opinion of the item under evaluation. For instance, SENGR [129] employs graphs to elucidate relationships between users and items and estimate aspects between each pair of interactions in the dataset. Similarly, SSG [56] aims to extract aspects from graph-based relationships, linking various users and items in interactions to make predictions about observed contexts.

With the defined main categories and subcategories, we classified all 124 articles selected in our mapping into the two categories and their respective subcategories. Table 2 presents the result of this classification. As evident, the classification is not entirely disjoint, meaning that works may be associated with more than one category. For instance, the AGTR algorithm [175] integrates Pairwise with emotion extraction to categorize the user-item relationship based on the emotion extracted from the comment. This approach identifies users who experienced the same emotion toward a particular item, deeming them similar preferences. Another example, Cheng et al. [28] extract global topics from comments—the closest topic represents each comment. Subsequently, a local extraction strategy employing attention mechanisms is employed, extracting the main topics to represent the user.

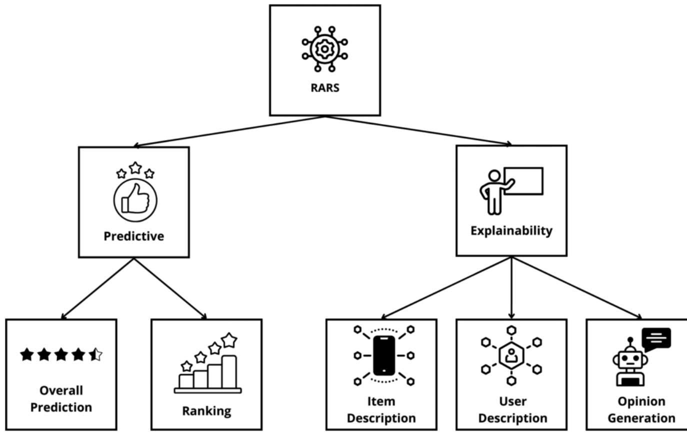

Fig. 5. Graphical representation of the observed objectives of the models.

3.3 Model Objectives

The third analysis perspective discusses the purpose of using the information extracted from the comments in the recommendation and explainability scenario. RARSs can cover different objectives in relation to their recommendation purpose. When using additional information to make recommendations, such as user comments, the systems have available a range of aspects and characteristics of users' implicit preferences, as well as implicit characteristics of items, which in the vast majority are not provided a priori. Armed with this contextual information, the proposed articles have the potential to furnish accurate recommendations and elucidate to users why certain items were suggested to them. This aspect of RSs falls within the domain of *Explainability* [1].

The collected articles can be divided into three objectives: (i) prediction systems, (ii) explanation systems, and (iii) hybrid systems encompassing both approaches. However, one of the filters applied while selecting relevant articles excludes those that solely utilize user comments to explain recommendations without introducing any innovations in recommendation techniques. Consequently, only prediction and hybrid systems were included in the final set of 124 articles. Figure 5 illustrates how we categorize the papers into their respective objectives. Details next.

3.3.1 Prediction Systems. RARSs classified into this category aim to predict ratings directly from comments, considering the numerical evaluations provided by users primarily while optimizing system parameters. These algorithms operate under the premise that comments serve as a rich and accurate source of user preferences, often surpassing the reliability of numerical ratings. Consequently, they seek to estimate a numerical rating directly from the comments to predict a new rating for the interaction or to balance existing user ratings for items.

For instance, McAuley et al. [110] utilize TM strategies to extract topics from user comments and items. They estimate a numerical rating by computing the average similarity between the topics discussed in user comments and those associated with items, indicating the user preference level for an item. Similarly, Wang et al. [145] employ TM for a collaborative recommendation, identifying similar users based on the topics covered in their comments.

With advances in deep learning, techniques have become more intricate and accurate. Zheng et al. [189] incorporate convolutional layers into neural networks to model comment sentences, obtaining contextual representations of users and items. These representations are concatenated for each user-item interaction pair and subjected to dimensionality reduction using factorization machines to estimate the user's preference for an item. Similarly, Yin et al. [174] employ convolutional layers

Table 3. Organization of Predictive and Hybrid RARSs According to Their Objectives

Predictive and Hybrids RARS	
Overall Prediction (85%)	**Ranking Prediction (26%)**
[3, 4, 7, 9, 12, 14, 15, 17, 21, 23, 28–31, 33, 40–43, 45, 46, 49, 50, 54–56, 58, 60, 63, 67, 70, 76–78, 81–83, 86, 88–90, 94–98, 100–104, 107, 112, 113, 113, 118, 120, 121, 126, 127, 129–131, 134, 135, 137, 138, 140, 146–148, 150, 151, 153, 154, 157, 158, 160, 161, 163–166, 170–175, 179, 180, 182–186, 189]	[5, 19, 20, 24, 26, 27, 51, 53, 59, 61, 61, 68, 71, 83, 92, 93, 109, 113, 115, 128, 138, 149, 167–169, 176]

to extract aspects from comments, utilizing attention mechanisms to identify relevant aspects associated with each comment. They use these aspects to represent users and items, performing dimensionality reduction to estimate user-item preferences.

A few other studies aim to balance the use of real numerical evaluations with those predicted from comments. Elahi et al. [51] employ BERT embeddings and dimensionality reduction to determine the sentiment implicit in user and item comments, weighting the original numerical ratings based on sentiment extraction. Similarly, Xv et al. [169] utilize ensemble training with multiple neural networks to reduce popularity biases in recommenders and unbiasedly weight numerical ratings. These approaches demonstrate the evolving complexity and effectiveness of RARSs in predicting user preferences directly from comments, often outperforming traditional numerical ratings as indicators of user affinity with items.

Both studies report good results. However, some authors [51, 176] argue that predicting and weighting numerical ratings can lead to biases in the formation of recommendation lists since very similar items can be identified as equally relevant to the user, thus generating redundant recommendation lists. As a result, the chance of the user leaving the system because of the lack of diversity in the recommendation lists is high. Creating recommendation lists is referred to in the field as *List-Wise* [99]. Faced with this problem, some RARS papers propose predicting the recommendation list directly and not just the regressed or weighted numerical evaluation. These algorithms are called **Ranking Prediction**, which aim to predict a ranking of items that the user could consume and evaluate the quality of this ranking so that redundant recommendations are minimized. RBLT [138], for instance, balances comments with numerical ratings and models latent topics from the resulting texts to recommend a list of top-N recommendations. Trirank [65] models user-item interactions using a bipartite graph and provides top-N recommendations according to the affinity balanced by the graph for different user interactions with various items.

As mentioned, algorithms can benefit from additional information gathered from comments to provide explanations for recommendations. That is because of the methods used to extract information from comments, which include various language processing techniques. TM aims to break down texts into latent factors known as topics, where a set of words represents each topic. Attention mechanisms use probability matrix techniques to identify keywords that can summarize the text's meaning. Other techniques, such as graph modeling, identify similar words and aspects in sets of items to group them according to their characteristics. Consequently, there are several ways of extracting this content from comments, and thus, each technique results in a set of words or latent factors used to infer the main aspects of the texts and the representations of users and items. Therefore, various studies utilize these resulting sets of words and latent factors to provide not only recommendation lists but also explanations of why items have been recommended, offering users insight into the implicit learning of the system.

As before, we classified the 124 papers into the two categories described above. Table 3 presents the result of this classification. As before, sub-classification is not entirely disjoint. For instance, in the article by Yang et al. [171], attention mechanisms are employed to identify keywords within comments and generate representations of users/items based on these keywords. Subsequently, the similarity between these representations is computed to estimate the user's rating for the item. Additionally, they implemented an auxiliary component to process the discrepancy between the predicted and actual ratings, thereby adjusting the user's other actual ratings based on the calculated error. The RBLT algorithm [138] is also classified under both Overall Prediction and Ranking Prediction. It utilizes global topic extraction to calculate the average of each user's favored topics and received items. Then, it computes the similarity between user topics and items to estimate a single final value. To ensure more accurate recommendations, it employs collaborative filters between user representations to generate recommendation lists. Furthermore, it provides recommendation lists concatenated with items exhibiting the highest topic similarity to the user.

3.3.2 Hybrid Systems. As previously discussed, numerous studies have introduced algorithms that offer recommendations and explanations, referred here as hybrid RARS. Broadly, many of these algorithms aim to identify keywords or attributes within sentences, allowing comments to be represented by latent factors of lower dimensionality compared to the original texts. Consequently, users and items can be characterized by latent aspects derived from the texts. Leveraging these representations, it becomes feasible to intersect the sets associated with users and items, thereby correlating users and items that share numerous common aspects. Besides the three categories related to prediction strategies outlined in the previous section, we discerned three distinct categories of articles based on how they furnish explanations for recommendations to the user. These strategies primarily differ according to the language processing techniques employed.

In the first category, labeled as **Item Description**, the primary objective is to provide insights into the aspects and contexts pivotal in recommending the items. In this category, it is typically assumed that the comments lack any intrinsic relationship with each other, with each comment reflecting distinct user preferences in their interactions. An exemplar of this category, ESCOFILT [121] employs extractive summarization techniques. Utilizing clustering techniques, ESCOFILT extracts and provides summaries of previous item comments as explanations. AENAR, another example [182], processes user and item identification information along with rating data. AENAR then offers recommendations and explanations grounded in aspects derived from user interactions.

A second category denoted as **User Description** posits that aggregating various comments into user-related documents makes it feasible to infer profiles or classes of user personalities within the dataset. These approaches assume that discernible patterns exist among user personalities, which can be distilled into discrete classes. Each class would signify a distinct personality type, enabling the classification of users according to these profiles. Automatic personality recognition aims to automatically identify such profiles. Many explanation strategies in RSs are rooted in these techniques to group users, offering explanations based on disparities between personalities. For instance, the JRL [185] model incorporates preferences into the attention generation process, ensuring that the generated explanations align with the preferences and sentiments expressed in the evaluations. Consequently, it becomes possible to discern users with convergent preferences and classify them.

Explainability through aspect and profile descriptions can yield significantly better results when compared to alternative explainability methods. However, they are limited to providing keywords or predetermined descriptions of user preferences. In real recommendation scenarios, where personalization is crucial for system success, furnishing preconceived explanations or static words does not guarantee the incorporation of implicit user preferences. Consequently, such approaches often yield overly general explanations. Yet, with the emergence of generative models and language generation techniques, numerous works in the literature have adopted these mechanisms within RARS.

Table 4. Organization of Hybrid RARSs According to Explainability Strategy

Explainability RARS		
Item Description (47.3%)	**User Description (21.2%)**	**Opinion Generation (31.5%)**
[21, 58, 60, 70, 87, 101, 104, 157, 182, 183, 185]	[57, 70, 85, 137, 159]	[7, 68, 115, 121, 165, 170, 175]

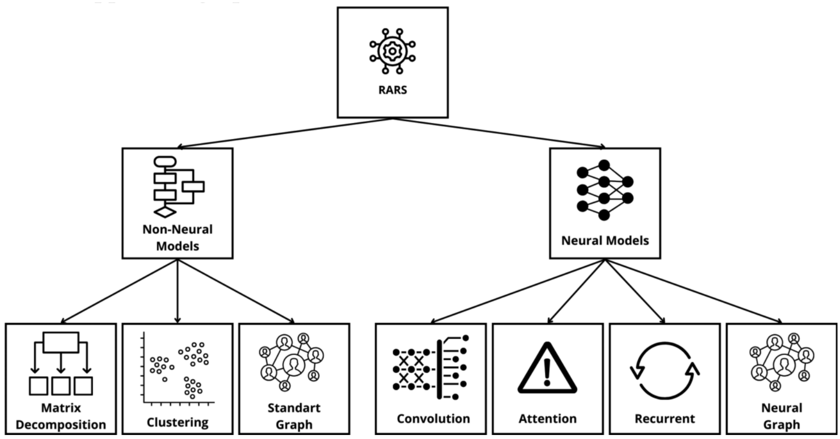

Fig. 6. Graphical representation of the ML architectures used by the papers.

Generative language models are neural models capable of encoding and decoding contexts with remarkable syntactic accuracy. Consequently, they can comprehend complex scenarios and provide increasingly personalized responses. Contemporary hybrid RARS systems have preferred leveraging generative language models as the primary explainability components. These models enable the generation of explanations for recommendations that are highly personalized and reliable, capturing implicit user preferences and item characteristics more effectively.

We categorize these works as **Opinion Generation**, aiming to extract various nuances from comments to construct dialogues regarding users' preferences and items' attributes. A prominent example is U-ARM [137], which utilizes attention learning of words extracted from user/item interactions to craft a textual sequence aligned with the recommendation. Another example is ExpansionNet [114], an advanced model designed to generate assessments by incorporating specific information about aspects. ExpansionNet initially uses a pre-trained ABAE model [64] to identify words associated with aspects. It associates users and items with representations that consider these aspects.

Among the 124 articles, we identified 20 deemed as hybrid, classifying them into the three categories mentioned above. Table 4 depicts the result of this (disjoint) classification.

3.4 Learning Architectures

The fourth and final perspective we analyze concerns the ML techniques employed to achieve the desired objective with the extracted information. Most of the collected papers utilize ANN architectures to process texts and perform regression on numerical evaluations, weights, or rankings alongside explainability techniques. The widespread adoption of ANNs for NLP [52], due to their effectiveness in these domains, has led the RARSs to intensify the use of these models [181].

We propose classifying the articles into two distinct categories for this perspective, as illustrated in Figure 6: (1) **Non-Neural Architectures:** these are founded on conventional learning techniques in the recommendation scenario, including matrix decomposition, clustering, and graph-based approaches; (2) **Neural Architectures:** these are constructed using ANN architectures, such as convolutional techniques, attention mechanisms, recurrent mechanisms, and neural networks applied to graphs. Below, we detail each of them.

3.4.1 Non-Neural Models. Non-neural models are distinguished for their efficiency in training. Due to their simpler computations and projections in latent spaces and reliance on relatively straightforward techniques compared to neural counterparts, non-neural models are generally more efficient and easier to implement. Among the various techniques discussed in this article, TM strategies emerge prominently. TM approaches are designed to extract and identify underlying "semantic topics" from raw textual documents [10, 47, 141]. In the context of RARS, TM algorithms have been extensively utilized in early proposals for representing users and items based on topics extracted from comments, with **Latent Dirichlet Allocation** (**LDA**) [10] being the most prevalent algorithm in this domain [6, 138].

Linear regression models have also been applied in the RARS scenario to estimate intrinsic aspects of texts. For instance, AGSRec [59] utilizes a graph model to estimate aspects of the relationships between users and items. Similarly, SUML [9] proposes a methodology employing a TM strategy to extract characteristics from comments. It estimates relevance coefficients for the extracted aspects from comments and users' numerical evaluations using regression. SUML also leverages MF for prediction. Other noteworthy non-neural models also focus on inferring sentiment in comments and assigning numerical ratings based on the sentiment expressed. Several studies have been proposed in this domain, including those by Anand et al. [3], Lei et al. [83], and Bauman et al. [9].

3.4.2 Neural Models. They dominate the current RARS landscape, being the most widely employed approach nowadays. Below, we will delve into the primary architectures utilized in this context.

3.4.2.1 Convolutional Networks: Models based on CNNs employ successive convolutional layers as key elements to extract implicit representations from texts. Ref. [78] introduced CNNs in RARS, where the representation of items was done by extracting features through convolutions of the comments obtained from interactions between users and items. DeepCoNN [189], built on the top of [78], uses two different convolutional networks to extract latent factors from both items and users. Similarly, Lin et al. [91] simplify the architecture of CNNs to capture contextual information from evaluations to improve the ability to represent characteristics of items and users. In addition, this model combines CNN and MF structures to boost recommendation performance. Kim et al. [79] used CNNs to capture contextual information from documents, considering local context and word order. RIGCN [13], CARM [86], TARERT [60], and NRPA [98] use convolutional layers to design attention structures and capture contextual information considering various types of contexts.

3.4.2.2 Attention Mechanisms: In NLP, attention mechanisms are strategies used to overcome the difficulties of conventional neural network models in representing long textual sequences. In the context of recommendation, we find several attention-based RSs. Attention mechanisms are the most widely used architectures within current RARS algorithms due to their effectiveness in modeling complex textual sequences. An example, NARRE [21] is a DNN model that uses attention to process relevant comments in the recommendation. DAMD [152] uses article descriptions in a multi-layered attention model to predict article characteristics, while [86] combines sequential information with layers of attention in order to project user behavior. Other models such as DAML [94], DATTN [126], MPCN [140], and TAERT [60] use global and local attention layers to infer evaluations of user-item pairs. *Dong et al.* and *Cheng et al.* models' [28, 49] seek to extract only the relevant signals from texts, selecting the most important aspects through layers of attention. NRPA [98] and TARMF [104] apply personalized attention models to learn specific representations of users and items based on their comments. Gao et al. [56] present a strategy to combine attention mechanisms with graph and sequential representations in the construction of RARSs.

3.4.2.3 Recurrent Neural Networks (RNNs): Recurrent models have gained prominence in processing temporal sequences. Faced with the problem of aggregating past and future information, RNNs sought to solve this issue by using internal states, where each state represented past iterations that could be merged with current information to predict future data. RNNs have been used for language processing since they can capture implicit factors in sequences of any kind. Cvejoski et al. [40] propose a dynamic neural network for processing comments. Given the user's sequence of comments, the system uses recursion to capture and store comments that have impacted user preferences. Kong et al. [81] use recursion in graph transitions to model interactions between users and items so that the most important interactions for the representations are passed on to future states through recursion.

3.4.2.4 Graph-Based Neural Networks: Recently, **graph-based neural networks** (**GNNs**) architectures have become popular in the recommendation scene. These architectures seek to integrate various techniques, such as attention mechanisms, convolutional networks, aspect and TM, with graph representation to build complex networks that capture user preferences and the distribution of items [162]. The work described in [4] uses a contrastive learning technique with comments to add interactions to the graph that did not exist before, alleviating the sparsity problem in the datasets. RI-GCN [13] models a bipartite graph of users and items, integrating comments and numerical ratings into the graph's relations and modeling the representations using convolutional layers to extract implicit contexts. SSG [56] takes a joint approach, modeling historical ratings in multiple perspectives, user-item sequences, and the bipartite user-item graph to improve efficient learning of user and item representations. Studies such as [103, 131, 156] employ hierarchical attention networks to extract features from reviews and attention-based models to obtain *embeddings* for nodes (users and items). Recently, graph-based convolutional network architectures [82, 146, 186] have proved effective in the RARS scenario. They can map contexts between user-item, user-user, and item-item relationships and project the representations into a contextual space capable of regressing implicit factors of the relationships. In [33], the authors present a strategy that captures user preferences and personalizes items from knowledge graphs, focusing on extracting sentiments from reviews and associating them with relationships in the graph, predicting the probability of recommendation based on cliques on the user graph. In [157], the authors propose a modular framework that, despite predicting user ratings, focuses on personalized explanations considering the inconsistency of sentiment between ratings and reviews, using encoders to separate these modalities into shared and private features. Jendal et al. [70] propose a framework composed of three processing steps in the comments: (i) representation of the review, (ii) aggregation of reviews into a global opinion vector, and (iii) combination of user and item vectors, to perform an explainable ranking. In [7], the authors put forward a hybrid model that uses enhanced embeddings to construct a user behavior graph. The model then propagates preferences with attention mechanisms and predicts ratings of new items while generating textual explanations of the recommendations. The model AXCF, presented in [77], classifies the words extracted from the comments into positive and negative keywords. After that, it calculates the probability of purchasing an item using a graph-based collaborative filtering strategy and generates a personalized recommendation list.

We have categorized the 124 articles into the two outlined categories. The outcome of this classification is presented in Table 5. Notably, in the case of predictive algorithms, the sub-classification is not entirely disjoint. For instance, Chen et al. [27] employ convolutional layers to reduce the dimensionality of user comments, followed by recursion mechanisms to sequentially process the sentences, ensuring interdependence among all words in the comments. SSG [56] utilizes bipartite

Table 5. Organization of RARSs According to Learning Architecture

Learning Architeture				
Non-Neural (22%)	**Convolution (51%)**	**Attention (56%)**	**Recurrent (9%)**	**Graph (20%)**
[3, 5, 9, 14, 19, 20, 23, 24, 26, 59, 61, 71, 83, 89, 92, 93, 109, 112, 118, 120, 128, 135, 138, 167, 176, 183, 184]	[4, 12, 13, 15, 17, 21, 31, 33, 43, 45, 46, 50, 56, 60, 63, 67, 68, 76, 78, 81, 82, 85, 86, 88, 94–98, 102–104, 107, 115, 126, 127, 129–131, 146, 148–151, 154, 158–161, 163, 168, 170–175, 179, 182, 185, 186, 189]	[4, 7, 15, 21, 27, 28, 30, 40–43, 49–51, 53, 54, 56, 57, 60, 68, 70, 76, 77, 81, 82, 85, 87, 88, 90, 94–98, 101–103, 107, 113, 121, 126, 127, 129–131, 134, 137, 146–150, 153, 154, 157, 159, 160, 163–166, 168, 171, 172, 174, 175, 179, 180, 182, 185, 186]	[27, 40, 41, 56, 81, 103, 140, 164, 166, 185]	[4, 7, 13, 33, 50, 56, 57, 59, 61, 63, 70, 77, 81, 82, 100–103, 113, 130, 131, 146, 153, 157, 182, 186]

Table 6. Organização dos RARSs de acordo com a abordagem de recomendação

Recommender Approach		
Content Based (52%)	**Collaborative Filtering (40%)**	**Hybrid (8%)**
[9, 12, 14, 15, 17, 20, 21, 23, 24, 26–29, 50, 51, 54, 57–59, 61, 63, 67, 71, 81, 83, 85, 89, 93–95, 98, 100, 109, 112, 115, 118, 118, 127–131, 137, 140, 146–148, 154, 158–160, 163, 165, 167, 170, 172, 175, 176, 182–184, 189]	[3–5, 7, 30, 31, 33, 40–43, 46, 49, 53, 55, 67, 68, 70, 77, 82, 87, 88, 90, 92, 96, 101–103, 107, 121, 126, 134, 135, 138, 149, 150, 153, 157, 161, 166, 168, 169, 173, 174, 179, 180, 186]	[19, 45, 56, 60, 76, 78, 86, 97, 104, 113, 151, 164, 171, 185]

graphs to represent user-item interactions, with one group of nodes representing users, another representing items, and the edges denoting comments provided by users to the items. Recursive layers are employed to model the comments, considering word inter-dependencies.At the end, attention mechanisms are applied to identify the main latent factors extracted from prior processing.

In Appendix D, we present in Table A2 a summary of the main characteristics of these learning architectures, highlighting their advantages, disadvantages, and applications.

3.5 Recommendation Approach

Finally, after having classified each considered RARSs relevant article according to: (i) pre-processing strategy, (ii) feature extraction technique, (iii) purpose of the models, and (iv) ML architecture, we proceed to consider the recommendation approach. In this last category, the RARSs systems can be classified into three different strategies: content-based [116], based on collaborative filters [125], and hybrid [11], which combines the two previous strategies. The distribution of studies across these categories can be seen in Table 6.

4 An Experimental Benchmark for RARSs

In Appendix A, we detail the SLR answering **RQ2** and **RQ3**. Our main conclusion in this analysis is a lack of scientific rigor in the experimental protocols and setup for most of the considered RARS studies. Indeed, most papers experimented and tested their algorithms with different metrics, datasets, and methodologies without sharing a common evaluation framework. As far as we know, no comparative study has been found aimed at covering distinct metrics to create a trustable and replicable benchmark for the field. Accordingly, here we propose *iRev*, a new RARS benchmark that may serve as a guideline for future research. It encompasses the main steps required in a solid experimental methodology:

Table 7. Overview of the Collections Used in the Evaluation

Dataset	# Users	# Items	Sparcity
Amazon - Digital Music	16.566	11.797	99.91%
Amazon - Musical Instruments	27.530	10.620	99.92%
Amazon - Office Products	101.501	27.965	99.92%
Yelp 2021 - Tucson	8.540	8.867	99,99%
Yelp 2021 - Tampa	18.437	8.664	99,99%
Yelp 2021 - Philadelphia	32.376	14.226	99.97%

(1) **Data selection and preparation:** where we describe the different datasets, highlighting their common characteristics, the main cleansing protocols, and the best practices to split the data into training and testing portions.
(2) **Evaluation metrics and statistical validation**: where we describe the desirable metrics to be applied in the RARS field and the type of statistical validation required to support any claims of superiority of one alternative over the others.
(3) **Baselines and parameters configuration**: where we describe relevant algorithms to compare according to the objectives of each new approach.

We found just one benchmarking framework similar to ours, the Neu-Review-Rec,[2] that is not as complete as ours. We perform a comparison between *iRev* and Neu-Review-Rec in Appendix C.

4.1 Data Selection and Preparation

Among all experimental proposals identified in recent papers, the majority of them explore common datasets from two applications: Amazon and Yelp. These datasets are available online and can be easily downloaded.[3] Each application has many datasets related to specific topics (class of items - Amazon or cities - Yelp). For our benchmark, we chose the three largest datasets from each application. In Table 7, we summarize the information from these six datasets.

We propose dividing the data into training, test, and validation sets in our benchmark based on the time associated with each user's log. The training set consists of the first 90% of the data from each user and the other 10% of logs are saved in the testing set. To identify the optimal parameters for all models, we select the last 10% of the user's data from the training set as a validation set. We chose this split due to replicability concerns as most of the analyzed RARS studies use this same split. All reviews are pre-processed by removing stopwords and applying lemmatization, strategies common to most of the articles analyzed.

4.2 Evaluation Metrics and Statistical Validation

To evaluate the algorithms' recommendations, we considered three different classes of metrics, which evaluate the algorithms from different perspectives. The first class involves error-based metrics (MSE, MAE, RMSE). The second class includes ranking-based metrics (Precision, Recall, **Normalized Discounted Cumulative Gain** (**NDCG**)). Diversity-based metrics—Diversity and Novelty—encompass the third class. Diversity metrics calculate the diversity of items based on the categories present in the recommendations. To compare the results achieved by different methods with statistical validation, considering different metrics, we adopt the Wilcoxon signed-rank test with a p-value = 0.05 [144]. This strategy has been applied in the literature due to the non-normal distribution of the recommendation datasets [67, 132, 155].

[2] https://github.com/ShomyLiu/Neu-Review-Rec
[3] Available at: https://www.kaggle.com/datasets/yelp-dataset/yelp-dataset and https://jmcauley.ucsd.edu/data/amazon/

4.3 Baselines and Parameters Configuration

To determine a set of algorithms to be compared with as potential baselines for the majority of future works in the field, we further examined the algorithms proposed in the 124 papers. We considered various aspects of the proposals to determine which ones would be included in our benchmark and, consequently, considered in our experimental evaluation. In our selection, we consider the number of citations as a criterion, but we go beyond that since older articles generally accumulate more citations over time. To get around this, we adopt a strategy similar to that presented in [155]. We divided the 124 selected articles into two groups: those published before 2019 and those published after 2019. For the articles before 2019, we consider the most cited, 10 of which stood out significantly in terms of citations and were selected to make up our benchmark. When selecting articles from 2019 onwards, we calculated a selection score that takes into account the following criteria:

— Number of algorithms used as a baseline in the experimental evaluation that is also present in the 10 most cited articles in group 1.
— Number of datasets and metrics used to evaluate the proposals to demonstrate their effectiveness in different scenarios.

Each score was normalized using min-max normalization and then added together to obtain the scores for each algorithm in the second group. We ordered the articles after 2019 according to these scores. From the fifth algorithm onwards, we noticed a large discrepancy in the scores, i.e., they use few baselines and evaluation metrics. Thus, we selected the five algorithms with the highest scores to make up our benchmark.

After a careful examination, we observed that none of the 15 algorithms initially chosen for the benchmark were sourced from articles published in 2023 or later. To ensure our experiment aligns with the latest advancements in the field, we included the highest-rated article from 2023 and 2024, determined by the previously calculated score.

Therefore, the final list of 16 algorithms chosen for the benchmark and experimental comparison is presented in Table 8. A detailed description of all of them is presented in Appendix B.

Codes were obtained from the respective authors' GitHub repositories whenever available. Links to these repositories can be found in the bibliography for easy reference. It is important to note that we made minimal modifications to these codes, such as adapting data input and calculating necessary metrics for result evaluation. The primary aim was to preserve the integrity of the original algorithms and ensure experimental replicability. In cases where the authors did not provide code, we implemented the algorithms based on their descriptions in the original papers.

Many selected articles lack explicit experimental configurations, leaving the parameter setting details unclear. While some authors provide codes online with configurations included, others do not offer such resources or detailed descriptions. Notably, the textual representations commonly employed for algorithm training are pre-trained semantic words representations like Word2Vec and FastText. None of the compared algorithms explored Transformer-based contextual representations [35]. To address this lack of uniformity, we conducted parameter tuning for each algorithm, as outlined in Table 9. The best alternative for each algorithm considering the list of options described in the Table was chosen, based on the performance in the validation set.

In the next section, we apply this experimental (replicable) protocol to summarize the results produced by these distinct algorithms. This benchmark is available at https://github.com/guibitten03/iRev. It is open and extensible – new datasets, metrics, and baselines can be incorporated over time.

5 Applying the Benchmark: Results and Discussions

Table 10 shows the results for all 16 selected RARS algorithms, considering six data collections (three from Amazon and three from Yelp) and eight evaluation metrics. For better visualization

Table 8. Quotes and Algorithm Scores

Paper Information					Data Modeling			Information Extraction					Learning Architecture					Model		
Method	Year	Venue	Citations & Score	Code	Document Modeling	Sentence Modeling	Rating Aggregation	Polarization	Emotion	Global Extraction	Local Extraction	Pair Wise	Non Neural	Convolution	Attention	Recurrent	Graph Neural	Overall Prediction	Ranking Prediction	Explainability
Before 2019																				
DeepCoNN [189]	2017	WSDM	1036	✓	✓						✓			✓				✓		
ConvMF [78]	2016	RecSys	841	✓	✓		✓			✓	✓			✓				✓		
NARRE [21]	2018	WWW	520	✓		✓					✓			✓	✓			✓		✓
D-ATTN [126]	2017	RecSys	475	✓	✓					✓	✓			✓	✓			✓		
MPCN [140]	2018	KDD	308	✓		✓					✓				✓			✓		
ALFM [29]	2018	WWW	291	✓	✓			✓		✓	✓		✓		✓			✓		
TransNet [17]	2017	RecSys	283	✓	✓	✓					✓	✓		✓				✓		
A3NCF [28]	2018	IJCAI	211	✓	✓					✓	✓		✓	✓	✓			✓		
TARMF [104]	2018	WWW	167			✓	✓			✓	✓				✓	✓		✓		
ANR [30]	2018	CIKM	158	✓	✓			✓			✓			✓	✓			✓		
After 2019																				
CARL [159]	2019	TOIS	1,917	✓	✓						✓			✓	✓			✓		
DAML [94]	2019	KDD	1.912	✓		✓					✓			✓	✓			✓		
CARP [85]	2019	SIGIR	1.905	✓	✓			✓			✓			✓	✓			✓		
CARM [86]	2021	Neuro computing	1.878	✓		✓	✓	✓			✓			✓				✓		
HRDR [97]	2020	Neuro computing	1.869	✓		✓	✓				✓			✓	✓			✓		
2023																				
MAN [171]	2023	AI - Springer	1.625			✓					✓	✓		✓	✓			✓		

Table 9. Algorithm Parameter Settings

Algorithm Parameter Configurations	
Parameters	**Values**
Dimensions of user and item representations	32
Word vector dimensions	300
Hidden state dimensions	32 a 256
Encoders technics used	TF-IDF [80], Word2Vec [111], GloVe [119] and FastText [72]
Dropout rate	0.5
Weight decay	$1e^{-3}$, $1e^{-4}$
Maximum document length	500 words
Learning rate	$1e^{-3}$, $2e^{-3}$
Number of convolutional filters	100
Window slide size	3, 4, 5
Batch size	32, 64 e 128
Training epochs	100
Loss function	MSE
Optimizer	ADAM [80], RMSprop [124]
Regularizers	L1, L2

Table 10. Experimental Results

Amazon																	
Dataset	Metric	ConvMF	DeepCoNN	TRANSNET	D-ATTN	NARRE	ALFM	TARMF	A3NCF	MPCN	ANR	CARL	DAML	CARP	HRDR	CARM	MAN
Digital Music	MSE	0.411	0.347	0.492	0.347	0.321	0.493	0.376	0.493	0.405	0.379	0.324	0.382	**0.302**	0.320	0.349	0.321
	MAE	0.452	0.384	0.474	0.347	0.345	0.479	0.461	0.472	0.421	0.373	0.336	0.403	**0.316**	0.336	0.371	0.334
	RMSE	0.634	0.583	0.694	0.583	0.559	0.694	0.608	0.694	0.630	0.608	0.562	**0.541**	**0.542**	0.559	0.583	0.566
	Precision@10	0.525	0.463	**0.799**	0.481	0.497	0.741	0.460	0.744	0.525	0.489	0.502	0.685	0.511	0.498	0.451	0.530
	Recall@10	0.274	0.319	0.224	0.304	0.345	0.224	0.339	0.204	0.261	0.286	0.331	**0.614**	0.366	0.333	0.366	0.398
	NDCG@10	0.000	0.000	0.000	0.024	0.016	0.016	**0.032**	0.024	**0.040**	0.024	0.016	0.000	0.024	**0.032**	**0.032**	**0.033**
	Diversity	0.050	0.045	0.053	0.052	0.041	0.053	0.052	0.053	0.053	**0.210**	0.055	**0.209**	0.088	0.051	0.117	0.079
	Novelty	0.071	0.061	0.053	0.043	0.043	0.013	0.155	0.012	0.048	0.049	0.046	**0.175**	0.041	0.040	0.044	0.044
Musical Instruments	MSE	0.927	0.884	0.977	0.889	0.862	**0.841**	0.882	0.977	0.922	0.903	**0.854**	0.925	0.884	0.860	0.885	0.894
	MAE	0.753	0.667	0.747	0.685	0.670	**0.637**	0.720	0.746	0.718	0.650	0.658	0.716	0.668	0.653	0.666	0.655
	RMSE	0.959	0.935	0.983	0.939	0.924	0.912	0.935	0.984	0.955	0.945	0.919	**0.887**	0.935	0.923	0.936	0.989
	Precision@10	0.507	0.442	0.364	0.402	0.414	**0.535**	0.386	0.154	0.509	0.497	0.455	0.491	0.434	0.417	0.385	**0.522**
	Recall@10	0.222	0.243	0.200	0.247	0.256	0.262	0.244	0.203	0.234	0.239	0.256	**0.386**	0.239	0.251	0.245	0.289
	NDCG@10	0.008	0.023	0.000	0.023	0.023	0.016	0.016	0.000	0.023	0.023	0.023	0.000	0.023	0.023	**0.031**	0.023
	Diversity	0.073	0.077	0.074	0.078	0.071	0.088	0.069	0.074	0.082	0.083	0.089	**0.302**	0.228	0.110	0.101	0.090
	Novelty	0.119	0.070	0.074	0.075	0.075	0.029	0.134	0.034	0.072	0.106	0.069	**0.281**	0.074	0.069	0.069	0.078
Office Products	MSE	0.984	0.933	1.028	0.928	0.922	1.028	0.922	1.028	0.970	0.946	0.943	1.002	0.930	**0.863**	0.927	0.886
	MAE	0.731	0.687	0.754	**0.664**	0.676	0.754	0.717	0.755	0.706	0.709	0.707	0.741	0.676	0.703	0.701	0.716
	RMSE	0.986	0.961	1.008	0.958	0.955	1.008	0.956	1.008	0.979	0.967	0.966	**0.921**	0.959	0.928	0.958	0.945
	Precision@10	0.474	0.399	0.152	0.385	0.379	0.152	0.371	0.152	0.458	0.466	**0.490**	0.461	0.373	0.391	0.337	0.410
	Recall@10	0.226	0.244	0.202	0.243	0.248	0.202	0.248	0.202	0.231	0.236	0.246	**0.350**	0.240	0.301	0.255	0.298
	NDCG@10	0.023	0.023	0.010	0.016	0.023	0.016	0.016	0.016	0.008	0.008	0.016	0.000	0.016	0.023	**0.039**	0.008
	Diversity	0.073	0.082	0.073	0.101	0.069	0.073	0.067	0.073	0.152	**0.353**	0.096	0.297	0.162	0.170	0.270	0.169
	Novelty	0.074	0.075	0.073	0.071	0.070	0.023	0.129	0.023	0.071	0.075	0.076	**0.286**	0.071	0.069	0.106	0.061

Yelp																	
Dataset	Metric	ConvMF	DeepCoNN	TRANSNET	D-ATTN	NARRE	ALFM	TARMF	A3NCF	MPCN	ANR	CARL	DAML	CARP	HRDR	CARM	MAN
Tucson	MSE	1.840	1.849	2.156	1.847	1.786	**1.700**	1.856	2.202	1.964	1.793	1.844	1.925	1.772	**1.698**	1.858	**1.702**
	MAE	1.144	1.078	1.239	1.073	1.060	1.057	1.081	1.239	1.162	1.091	1.107	1.168	1.078	1.057	1.089	**1.001**
	RMSE	1.355	1.357	1.467	1.356	1.333	**1.301**	1.359	1.482	1.399	1.337	1.355	1.359	1.328	**1.302**	1.360	**1.299**
	Precision@10	0.321	0.349	0.134	0.354	0.365	0.379	**0.380**	0.206	0.312	0.355	0.367	0.235	0.337	0.337	0.362	0.346
	Recall@10	0.237	0.272	0.208	0.265	0.275	0.252	**0.286**	0.203	0.231	0.259	0.271	0.237	0.265	0.234	**0.280**	0.190
	NDCG@10	0.000	0.000	0.006	0.010	**0.038**	0.009	0.000	0.008	0.008	0.000	0.015	0.000	0.015	0.010	0.008	0.010
	Diversity	0.312	0.253	0.348	0.409	0.202	0.255	0.280	0.120	0.360	0.141	0.139	**0.475**	0.130	0.219	0.210	0.122
	Novelty	0.139	0.102	0.111	0.120	0.100	0.006	0.134	0.008	0.104	0.095	0.155	**0.392**	0.097	0.110	0.133	0.095
Tampa	MSE	1.647	1.696	1.756	1.715	1.631	**1.533**	1.702	1.962	1.791	1.657	1.720	1.737	1.600	1.695	1.710	1.629
	MAE	1.056	1.008	1.082	1.038	0.988	0.997	1.007	1.184	1.081	1.002	1.041	1.088	**0.977**	1.003	1.021	1.001
	RMSE	1.281	1.299	1.323	1.307	1.274	**1.236**	1.302	1.399	1.336	1.284	1.309	1.285	1.262	1.299	1.305	1.276
	Precision@10	0.347	0.359	0.288	0.379	0.372	0.313	**0.384**	0.230	0.322	0.318	0.318	0.266	0.330	0.310	**0.380**	0.295
	Recall@10	0.239	0.276	0.222	0.280	0.276	0.255	0.284	0.200	0.235	0.251	0.253	0.261	0.262	0.274	0.272	**0.304**
	NDCG@10	0.019	0.016	0.000	0.016	**0.038**	0.010	0.016	0.018	0.010	0.016	0.016	0.000	0.016	0.013	0.008	0.011
	Diversity	0.045	0.120	0.120	0.036	0.133	0.103	0.085	0.067	0.082	**0.192**	0.116	0.112	0.184	0.109	0.087	0.111
	Novelty	0.141	0.094	0.095	0.156	0.093	0.021	0.103	0.020	0.107	0.152	0.096	**0.384**	0.140	0.090	0.105	0.095
Philadelphia	MSE	1.416	1.487	1.611	1.473	1.412	**1.372**	1.479	1.611	1.586	1.450	1.475	1.545	1.401	1.451	1.450	1.420
	MAE	0.952	0.952	1.076	0.941	**0.923**	**0.926**	0.932	1.075	0.995	0.932	0.951	1.010	0.933	0.952	**0.923**	0.976
	RMSE	1.188	1.217	1.206	1.211	1.185	**1.169**	1.213	1.206	1.256	1.201	1.212	1.209	1.181	1.210	1.201	1.180
	Precision@10	0.346	0.358	0.290	0.364	0.365	**0.400**	0.377	0.301	0.322	0.340	0.339	0.286	0.335	0.310	0.363	0.358
	Recall@10	0.242	0.261	0.200	0.254	0.267	0.255	**0.272**	0.200	0.225	0.249	0.247	**0.275**	0.262	0.262	0.268	0.260
	NDCG@10	0.000	0.001	0.013	0.019	0.000	0.012	0.013	0.002	0.012	0.002	0.014	0.019	0.018	**0.027**	0.019	**0.026**
	Diversity	0.084	0.085	0.098	0.097	0.084	0.091	0.092	0.137	0.095	**0.350**	0.083	0.117	0.222	0.272	0.110	0.137
	Novelty	0.084	0.085	0.098	0.097	0.084	0.049	0.091	0.018	0.092	0.137	0.095	**0.350**	0.083	0.097	0.110	0.091

and interpretation of the results, the algorithms in the header have been arranged in order of publication, with the leftmost corresponding to older works and the rightmost to more recent ones. Results with gray cells were the best for a metric in a given dataset. The rows containing more than one gray cell consist of algorithms that tied statistically as the best results (Wilcoxon with p-value equal to 0.05).

For the ranking error metrics, such as precision, recall, and Ndcg, we used recommendation lists containing 10 items (Precision@10, Recall@10, and Ndcg@10), as discussed in [75] in their systematic mapping on the use of texts in RSs. Most of the best results are on the right-hand side of the table, demonstrating the evolution of the algorithms' effectiveness over time since most recent systems outperform the older ones. Next, we analyze the results from three different perspectives, according to the metric classes presented above.

5.1 Error Evaluation

Most of the algorithms' results obtained in our experiments are consistent with those reported in the original works, statistically beating the corresponding baselines to which they were originally compared. In general, NARRE, ALFM, DAML, and CARP stood out regarding error metrics. Most of them use local feature extraction in their systems, while ALFM uses global extraction. We can deduce that the extraction of local features allows for a specific but still generalist extraction and consequently manages to cater to the majority of users, indicated by the good MSE and RMSE results, which penalize, on average, predicted values that are very discrepant from the real values.

Along the same lines, similarly to ALMF, CARP results are promising. CARP performs local feature extraction and then introduces a mechanism called capsules—groups or sets of extracted polarized features—so that the system recognizes positive, negative, and neutral capsules and assigns them to interactions. Despite not obtaining the best values for some scenarios, CARP's results were very close to the best ones, contrasting with its predecessor, CARL, which does not apply sentiment extraction and obtained worse results.

NARRE did not obtain the best value for any error metric in the data collections, but it always remained in the top five in all datasets. Indeed, NARRE was the most consistent method when considering the set of all error metrics in all datasets.

Finally, Transnet and A3NCF obtained the worst performances among the error metrics. A3NCF, despite using document modeling and global feature extraction as ALFM. A3NCF does not have the bias evaluation mechanisms present in the ALFM, which may have been a determining factor in the good performance of the latter.

5.2 Ranking Evaluation

Concerning the ranking metrics, we observe that, in general, the values obtained for NDCG are very low compared to those of the other tested metrics. This occurs because most users do not have many interactions on the dataset, as shown in Table 7, resulting in a high sparsity. Because of this, and the separation of the dataset between training, testing, and validation, the average size of the list of items consumed by the user for testing the algorithms drops dramatically. Most of the time, the test lists contained around 1 to 2 relevant items. This may also indicate that the cutoff point 10, used by most works that employed NDCG, may not be ideal for such a task. Lower values for the metric (@2, @3 or @5) or metrics such as **Mean Reciprocal Rank** (**MRR**) may be better suited.

In any case, it is still possible to see that some algorithms, such as CARM and TARMF, stood out. Indeed, we observed that algorithms that use rating aggregations in text modeling obtained excellent results among the ranking metrics. CARM, which uses only convolutional layers and rating aggregations, obtained the best NDCG results for all of Amazon's datasets and consistent results for the same metric for Yelp's. TARMF, which uses attention and recurrence mechanisms, along with rating aggregations, obtained the best precision and recall results for several Yelp datasets while obtaining good NDCG results for Amazon datasets.

We also found that for different scenarios, the best algorithms vary considerably. For instance, the recently proposed MAN algorithm (2023) did not obtain the best results for most datasets, but, in general, this algorithm was the most consistent across different scenarios. Similarly, NARRE, which did not stand out as the best method in most scenarios, remains among the top five algorithms for many scenarios, obtaining consistent ranking results for all datasets. DAML was very effective in precision and recall for Amazon datasets, obtaining the best recall results for all data collections. However, the same system obtained average results for the Yelp datasets.

All methods mentioned above use sentence modeling, which allows extracting in a more detailed way the user's preferences and items' characteristics from the sets of comments provided by users and more personalized recommendations, possibly explaining the good results achieved.

5.3 Diversity Evaluation

The third and final analysis concerns the diversity perspective (diversity and novelty metrics). Intuitively, it is reasonable to think that maximizing diversity measures for users with few preferences may result in worse ranking measures since the recommendation lists will be more diverse and less specific (and probably contain less relevant items) [133]. Despite that, we observe that certain algorithms that obtained good results in error and ranking metrics were not necessarily bad in the diversity metrics. A notable example is the DAML algorithm, which achieves the best accuracy and ranking by exploring different categories in the Amazon datasets. Such exploration makes this algorithm one of the best in terms of diversity as well.

On the other hand, in datasets from Yelp, where there is a natural bias from the geographical distances from one store to another in the cities, algorithms with the best results in accuracy and ranking are not necessarily the best ones in diversity. As users tend to consume items close to their locations, the algorithms may have identified more implicit patterns in user consumption and have thus personalized the recommendations better. Such a level of personalization may result in a lack of diversity overall. Besides these factors, the diversity is also related to the selected approach of each algorithm. The ANR algorithm, which combines global and local feature extraction to remove popularity biases, can also achieve excellent diversity results compared to others.

As for novelty, algorithms that usually achieve more diversity have a higher probability of presenting unexpected items to the user. In this sense, experimental results also highlight the DAML as a good option to maximize novelty for our users. Moreover, the novelty metric is calculated by measuring how unpopular the items that have been recommended are. In this case, algorithms that employ topic extraction, such as ALFM and A3NCF, do not obtain good results in novelty since the characteristics obtained globally are usually the characteristics extracted from the most popular items, and therefore, the recommendation has a popularity bias.

5.4 Final Discussion

To summarize the above discussions, we evaluate all the models using a unified perspective based on a metric adapted from Game Theory, the Multi-Attribute Utility Metric (MAUT) [16]. MAUT aims to represent the performance of a method as a function of all M quality metrics, where $m_{i,j}$ is the utility of metric i for method j. For effectiveness metrics, the higher the value, the better the method. For error metrics, the lower the value, the better the method. Therefore, $m_{i,j}$ can be mapped directly to the value obtained by method j in metric i. For each metric, the usefulness of a method is calculated using min-max normalization. This normalization makes it possible to rank the methods of a given metric from best ($U_{i,j} = 1$) to worst ($U_{i,j} = 0$). The utility function of a method U_j can be defined in the form of an additive model as: $U_j = \sum_{i=1}^{|M|} U_{i,j} \cdot w_i$, where w_i

Table 11. Sum of the MAUT Metrics Calculated for the Metric Classes in Data Collections

Method	Amazon Datasets				Yelp Datasets				All datasets
	MAUT				MAUT				∑
	Error	Ranking	Diversity	Sum	Error	Ranking	Diversity	Sum	(Sum)
CARP (2019)	6.75	4.5	3.56	14.81	7.62	5.43	3.25	16.3	31.1
MAN (2023)	6.0	6.62	3.62	16.24	7.37	5.0	2.43	14.8	31.04
HRDR (2020)	7.75	5.75	2.62	16.12	5.87	4.81	3.37	14.05	30.17
CARM (2021)	5.56	5.43	4.12	15.11	4.56	6.25	3.68	14.49	29.6
NARRE (2018)	5.56	1.56	2.5	9.62	7.62	7.0	2.25	16.87	26.49
ANR (2018)	4.5	4.75	1.37	10.62	6.06	3.93	4.37	14.36	24.98
CARL (2019)	5.43	3.56	2.37	11.36	3.68	5.31	3.06	12.05	23.41
ALFM (2018)	4.5	1.75	0.31	6.56	8.56	5.12	1.68	15.36	21.92
DAML (2019)	5.5	5.87	1.81	13.18	2.62	3.18	5.31	11.11	24.19
D-ATTN (2017)	4.87	2.93	1.87	9.67	4.31	6.75	3.87	14.93	24.6
TARMF (2018)	4.31	3.25	1.18	8.74	4.25	7.18	3.25	14.68	23.42
DeepCoNN (2017)	4.87	2.87	1.81	9.55	4.12	5.5	2.75	12.37	21.92
ConvMF (2016)	4.62	3.12	0.81	8.55	5.43	3.87	3.06	12.36	20.91
MPCN (2018)	5.06	3.06	1.0	9.12	1.62	2.87	3.31	7.8	16.92
TRANSNET (2017)	2.37	3.12	0.5	5.99	1.68	1.87	4.0	7.55	13.54
A3NCF (2018)	2.56	1.18	0.31	4.05	1.06	2.37	1.31	4.74	8.79

represents the weight of metric i and usually satisfies the condition $\sum_{i=1}^{|M|} w_i = 1$. For precision (Precison, Recall, and NDCG) and diversity (Diversity and Novelty) metrics, the most effective RARS strategies are those that result in the highest values. The exception is error metrics (MSE, MAE, and RMSE), where the lower the value, the more effective the strategy. For the sake of standardization in the calculation of min-max normalization, we multiplied the values of these metrics by -1.

We assume all metrics are independent and equally important in the decision process. Thus, the higher the score, the better the algorithm. The MAUT metric was calculated for each class of metrics for each collection of data and then added up for each scenario separately (Amazon and Yelp - column "Sum") and aggregate (column "All datasets ∑ Sum"), which can be seen in Table 11. Results with gray cells were the best for a set of metrics in a given scenario.

As we can see, no RARS excelled in all scenarios. In other words, no algorithm stood out to the extent of achieving hegemony in any class of metric. The most effective RARS in one class of metrics obtained poor or average results in other classes. Indeed, none of them consistently achieved the best results for more than three sets of metrics. For instance, the best RARS for error metrics in the Amazon datasets is the HRDR, which achieves only average results in Yelp.

Analyzing the three sets of metrics together, we have that the best RARSs for the Amazon datasets are MAN (2023), HRDR (2020), and CARM (2021). On the other hand, for the Yelp datasets, the best RARSs are completely different: NARRE (2018), CARP (2019), and ALFM (2018). Considering all scenarios and datasets, the highlights are CARP, MAN, and HRDR from 2019, 2023, and 2020, respectively, which are not always considered baselines in the most recent work. All these results and analyses demonstrate the need for more comprehensive experimentation concerning the used metrics and the evaluated scenarios. These results also reaffirm the importance of a comparative benchmark framework like the one we propose in this article. We complement these analyses by comparing the five best RARS algorithms with five traditional RS. The results are detailed in Appendix E.

6 Future Research Directions

Our work leads the research direction toward a more pronounced concern with reproducibility. This entails a more thorough and equitable comparison of new proposals, encompassing a diverse array of data collections, a broader spectrum of metrics that consider various aspects, and a more significant number of baselines, all while rigorously validating the results statistically. Furthermore, there is an increasing demand for the public availability of the codes used. In this context, our framework stands out as an example of a thorough research process to be followed.

Our investigation reveals room for multi-objective review-aware recommendation proposals that simultaneously maximize several recommendation quality criteria (not only effectiveness). Although contextual embeddings present excellent results in other research fields [44], none of the investigated works directly adopt them in their proposals, which opens up opportunities.

Upon analyzing the results of our experimental evaluation, it becomes evident that strategies relying solely on convolutional networks or TM [17, 28, 78, 140, 189] are outperformed by those that integrate more than one type of technique and/or architecture. The most promising outcomes are achieved by combining convolutional layers and attention mechanisms [85, 86, 97, 171]. These findings underscore the significance of exploring the combination of different neural architectures and complementary techniques, paving the way for more impactful research in RARs.

Although it was not the focus of the present research, explainability in RSs was a topic that appeared with some frequency in our SLR [48, 114]. On the other hand, more recently, LLMs have become important tools to assist in the explainability task [84, 105, 188]. In this sense, due to the intrinsic characteristics of RARs in the use of text, there is a natural adherence to LLMs. Therefore, we see it as a very promising research direction.

Finally, we observe an increasing number of RS studies related to the exploration of reinforcement learning, mainly based on multi-armed bandits [132]. These works argue that users' preferences change over time, and the RSs must be able to adapt themselves. Despite the growth of reviews written by users and their potential to elucidate users' preferences, we have not (yet) observed much effort on employing reinforcement learning with this goal.

7 Conclusions

This article presented an SLR of the RARS realm to provide a summarization of the current state-of-the-art and open challenges. The last similar endeavor was reported nine years ago, well before the revolution of NLP and related tasks due to the DNNs revolution with word embeddings, attention models, transformer architectures, and so on. Our current work, focused on recent RARSs proposals, captures such a revolution.

By inspecting 832 articles published in the last 10 years (2014–2023), we identified 124 articles as the most relevant RARSs studies. We proposed a new taxonomy regarding how recommendation algorithms use NLP techniques to define user preferences through their comments and use the taxonomy to categorize the 124 papers under various and complementary perspectives. Our taxonomy classifies the articles into four primary perspectives: (i) Data Modeling (how textual data are pre-processed and modeled); (ii) Information Extraction (how user preferences and item characteristics are extracted from textual data); (iii) Model Purpose (how the textual data is used—to predict, to explain, or both); and Learning Architecture (which ML techniques are used during the creation of the recommendation model). To our knowledge, it is the first article that proposes a detailed organization for the RARSs literature.

We also analyzed all these papers from the point of view of the scientific rigor and criteria with which the algorithmic proposal of these works has been scrutinized, including issues regarding parameterization, datasets, and evaluation metrics. We observed that most of the selected studies evaluate a single quality dimension, effectiveness, despite the consensus in the RS community that other quality dimensions (e.g., novelty and diversity) are essential to assess the practicality of recommendations.

Regarding the adopted experimental setup, we observed a low intersection of metrics and datasets used in the experiments, and a few distinct studies use many metrics. More importantly, there was a lack of rigorous comparative studies among the existing proposals under the same common setup. All of this establishes the need for a common experimental framework that guarantees the reproducibility of evaluations and straightforward comparison of results, considering different and complementary quality dimensions (accuracy, novelty, and diversity).

Accordingly, we proposed and evaluated a RARS benchmark to be considered in future evaluations. Our benchmark initially considers 16 different baseline algorithms, six datasets, and six evaluation metrics. Other datasets, metrics, and baseline algorithms can be easily incorporated in the future. We evaluated our proposed benchmark by performing experiments and comparing the implemented algorithms. Our results demonstrated that identifying a "single best" algorithm is not a trivial task as there is no single algorithm that performs consistently well across all investigated scenarios.

We will keep complementing the created repository with (new) strategies, making it a reference for researchers in the area to use in their evaluation experiments and to contribute with the source code of their respective solutions.

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Appendices

A Systematic Evaluation

After addressing **RQ1** through a comprehensive literature analysis, categorizing the selected papers according to the taxonomy proposed, we proceed to tackle the subsequent two research questions: **RQ2:** What are the predominant datasets and evaluation metrics utilized in the RARSs domain? **RQ3:** How are experimental evaluations conducted in the scrutinized studies, considering the current state-of-the-art configurations and model parameters? To this end, we conducted a comprehensive characterization of the 124 papers, as outlined in Table A1, encompassing information such as total citation count, publication year, datasets employed in experimental evaluations, utilized metrics, and availability of source code. Subsequent sections provide a detailed analysis of the information in the table.

A.1 Datasets

We conducted a count of all datasets employed in the experimental evaluations of the 124 RASRs papers distributed in Table A1. The distribution of these datasets is depicted in Figure 7.

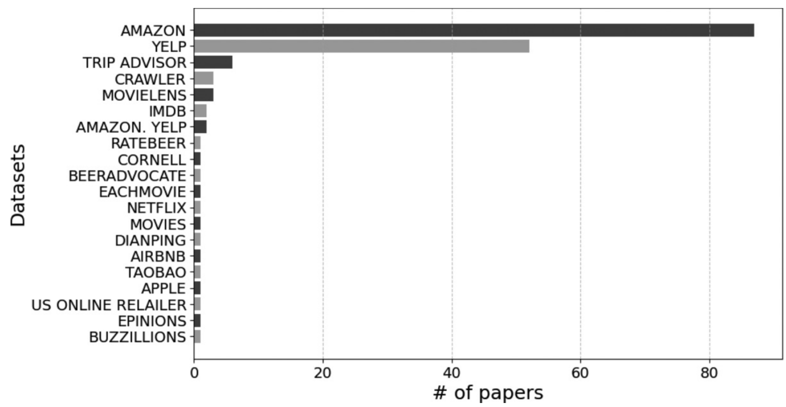

Fig. 7. Frequency of datasets observed in the search and collection.

It is evident from our analysis that the Amazon dataset, encompassing a wide array of products ranging from books to music and movies, alongside the Yelp dataset, which pertains to establishments, restaurants, and tourist attractions primarily within the United States (US), emerge as the predominant datasets utilized in the RARSs context. They exhibit a notable disparity when compared to the third most utilized dataset onward. These datasets hold significance in review-aware recommendation analyses owing to the substantial volume of user-generated comments associated with the items.

We conduct a more in-depth examination of the two most prevalent datasets, namely Amazon and Yelp, which feature various categories and versions. To support a nuanced analysis of the categories within Amazon and versions of Yelp most frequently employed in the compiled works, we analyzed the utilization of each category within the two datasets, aiming to pinpoint the predominant scenarios explored in the literature. Figures 8 and 9 depict the results of this analysis.

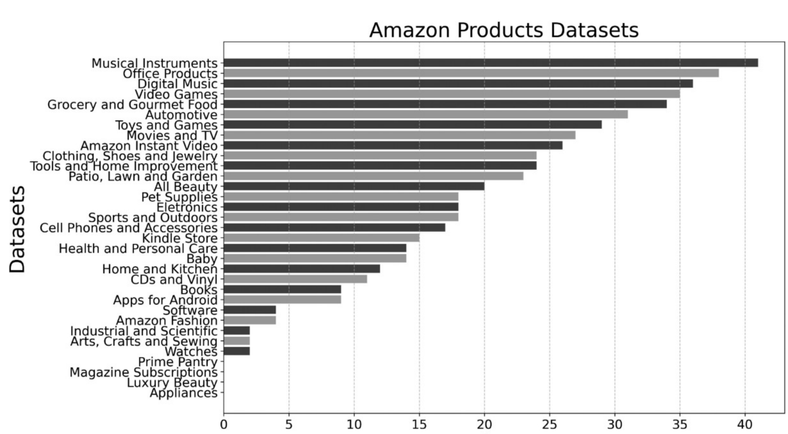

Fig. 8. Frequency of each category in Amazon's datasets.

In the case of Amazon, as illustrated in Figure 8, the prevalence of specific categories for utilization in experimental evaluations correlates with the popularity of the items sold within those categories and, consequently, with the total number of available reviews. Notably, categories such as Musical Instruments, Digital Music, and Office Products emerge as popular choices. Conversely, categories like Industrial and Scientific are among the least utilized.

Regarding Yelp, the situation is considerably more intricate. The dataset is segmented by city; however, a significant proportion of entries lack specification regarding the city utilized in the experiments, posing a challenge for reproducibility and comparability across studies. Additionally, *Yelp* encompasses time slices, as depicted in Figure 9, with experimental evaluations often employing disparate time slices. These issues present obstacles to the reproducibility of studies utilizing Yelp data.

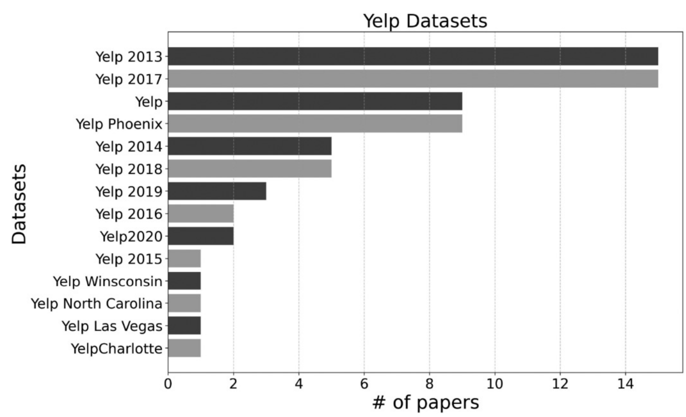

Fig. 9. Frequency of each year and category in the Yelp dataset.

A.2 Metrics

After examining Table 10, we accounted for all metrics employed in the experimental evaluations of the 124 RASRs papers. Figure 10 depicts the distribution of these metrics.

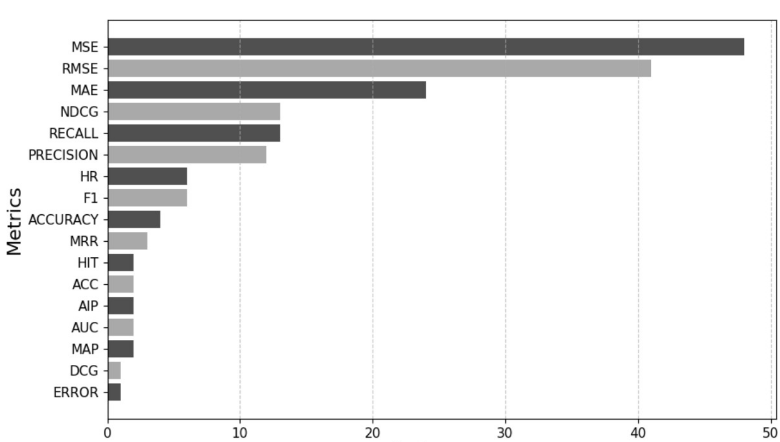

Fig. 10. Frequency of metrics observed in search and collection.

Regarding metrics focused on evaluating the recommendation's quality, the three most commonly utilized ones are error metrics. *MSE* (Equation (1)) holds prominence, serving as a reliable indicator of model accuracy. It accords greater importance to more significant errors by squaring each error individually before computing their average. Similarly, *RMSE* metric (Equation (2)) serves as an absolute error measure, wherein deviations are squared to prevent positive and negative deviations from nullifying each other. Lastly, *MAE* (Equation (3)) adopts a linear approach to assigning weights to differences, allocating equal weight to all deviations irrespective of magnitude.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3}$$

The remaining metrics in experimental evaluations are ranking-based: NDCG, Recall, and Precision. *NDCG* (Equation (4)) measures the quality of a list of results returned by the model concerning the items that the user indicated as relevant. The assumption is that ranking relevant items in the first positions may produce more impact on the quality of recommendations. *Recall* (Equation (7)) is commonly used in a situation where False Negatives are considered more harmful than False Positives. *Precision* (Equation (8)) is commonly used in a situation where False Positives are considered more harmful than False Negatives.

$$NDCG@k = \frac{DCG@k}{IDCG@k} \tag{4}$$

$$DCG@k = \sum_{i=1}^{k}\frac{2^{relevant_i} - 1}{\log_2(i+1)} \tag{5}$$

$$IDCG@k = \sum_{i=1}^{k_{\text{truth}}}\frac{2^{relevant_{\text{truth}_i}} - 1}{\log_2(i+1)} \tag{6}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \tag{7}$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \tag{8}$$

However, besides the consensus within the RS community that accuracy alone is insufficient for evaluating the practical added value of recommendations [133], our analysis reinforces the persistent focus on such error metrics. Other evaluation perspectives, such as *diversity* and *novelty*, have not been addressed by any of the 124 selected papers. In the recommendation field, *Diversity* (Equation (9)) is used to measure the degree of dissimilarity among the N recommended items by evaluating their characteristics, which may include category distribution, latent factors, or other relevant features. In turn, *Novelty* (Equation (10)) measures how unexpected the items are for each user by calculating the inverse of the normalized popularity of each item in the recommendation list. Contrasting all papers identified in our SLR, both metrics are included in our experimental evaluation in the next sections.

$$\text{Diversity} = \sum_{i \in N}\sum_{j \in N, i \neq j} 1 - cossim(features(i), features(j)) \tag{9}$$

$$\text{Novelty} = 1 - \sum_{i \in N} \frac{log_2(popularity(i))}{|N|} \tag{10}$$

A.3 Algorithms and Parameters Configuration

Our third analysis of the RARSs' experimental environments considers the used algorithms. Our first observation concerns the availability of the source code of the proposed algorithms. Table A1, column "code", shows that less than 50% of the papers make the source code of their proposals available. This severely harms their use as baselines in new proposals. It also directly impacts the number of algorithms considered as baselines in these experimental evaluations. As we can see in Table A1 (column "Baselines"), a new algorithm is compared, on average, with only three other algorithms and, at most, nine other algorithms. That directly impacts the assessment of the real impact of a new approach on the advancement of the field.

Another important observation concerns the documentation of the calibration process of the algorithms used in the experimentation (baselines and new proposal). As seen in Table A1 (column "Parameters"), less than 30% of the selected papers detail the parameter calibration processes of the considered algorithms. We can see that the vast majority of papers that provide algorithm parameter specifications and training details are non-neural algorithms, which directly impacts the replicability of papers that propose innovations using neural architectures since parameter calibration is an essential process, especially regarding neural alternatives. All these aspects reinforce the existing difficulties in achieving satisfactory levels of reproducibility in the RARSs scenario.

Table A1. Detailed Characterization of the Algorithms Selected by SLR

Name	Citations	Year	Datasets	Metrics	Code	Baselines	Parameters
MCCL [153]	3	2024	Amazon	Ndcg, Recall		0	✓
RAKCR [33]	3	2024	Movielens, Amazon, Yelp	F1, Auc		0	✓
PESI [157]	2	2024	Yelp, TripAdvisor, Amazon	Rmse, Mae		0	✓
HypAR [70]	2	2024	Amazon	Auc, Map, Ndcg, Precision, Recall, F1, Div	✓	2	
KEGC [7]	0	2024	Amazon	Rmse, Mae	✓	0	
AXCF [77]	0	2024	SemEval, yelp	Precision, Recall, F1		0	
KAAN [101]	6	2023	Imdb, Amazon	Rmse, Acc, Auc	✓	3	
ARPH [82]	0	2023	Amazon	Rmse, Mae		5	
AHOR [146]	0	2023	Amazon, Yelp	Hr,Ndcg		3	
GCN [186]	2	2023	Amazon, Epinions	Rmse, Mae		5	
DMIR [172]	0	2023	Amazon	Mse	✓	6	
MAN [171]	3	2023	Amazon	Rmse, Mae		9	
BERT Sentiment [51]	1	2023	Amazon	Hit, Precision		3	
EGCF [4]	13	2022	Amazon	Recall, Ndcg, MAR		1	✓
DeepIDRS [68]	18	2022	Amazon	Hit, Ndcg		3	
ACNNDS [174]	1	2022	Amazon	Rmse, Mae		7	
CISD [63]	1	2022	Amazon, Yelp	Mse	✓	3	
RI-GCN [13]	5	2022	Amazon, Yelp	Mse		3	
AENAR [182]	3	2022	Amazon	Rmse		2	
Dualgcn [130]	7	2022	Amazon, Yelp	Ndcg, Aip, Mae, Precision, Recall, F1	✓	3	
DRR [40]	4	2022	Amazon	Mse		2	✓
LUME [169]	0	2022	Amazon, Yelp	Mse, Ndcg		4	

(Continued)

Table A1. Continued

Name	Citations	Year	Datasets	Metrics	Code	Baselines	Parameters
RNCDR [100]	13	2022	Amazon	Hr, Recall, Ndcg		2	
RGCL [131]	17	2022	Amazon, Yelp	Mse	✓	3	
KMRR [102]	2	2022	Imdb, Amazon	Rmse		2	
BanditProp [149]	1	2022	Amazon, Yelp	Precision, Recall, Map, Ndcg		2	
SentiAttn [150]	1	2022	Amazon, Yelp	Rmse, Mae	✓	4	
PSAR [147]	0	2022	Amazon	Ndcg, Hr		2	
SENGR [129]	5	2022	Amazon, Yelp	Mse, Aip		4	
SER [31]	0	2022	Amazon, Yelp	Mse, Ndcg		2	
GVN [161]	0	2022	Amazon	F1, Hr, Ndcg	✓	2	
GAR [81]	0	2022	Amazon, Yelp	Mae		2	
R3 [115]	4	2022	Amazon, Yelp	Mse, Pcc		4	
TAERT [60]	14	2021	Amazon	Rmse, Mae	✓	4	
ESCOFILT [121]	16	2021	Amazon	Rmse	✓	2	
SIFN [180]	6	2021	Amazon	Mse	✓	4	
AARM [58]	102	2021	Amazon	Recall, Precision, Ndcg, Ht	✓	3	
ADRS [43]	31	2021	Amazon, Yelp	Rmse, Mse		4	
HACN [50]	11	2021	Amazon	Rmse, Mae		2	
RPRM [148]	17	2021	Amazon, Yelp	Precision, Recall, Map		2	
ERP [160]	3	2021	Amazon, Yelp	Mse, Rmse		3	
JRL [185]	15	2021	Amazon	Mse, Mae	✓	2	
CARM [86]	41	2021	Amazon, Yelp	Rmse, Mae	✓	2	✓
RRNet [170]	0	2021	Amazon	Rmse	✓	4	
TGAT [103]	22	2021	Amazon, Yelp	Mse	✓	2	
DRRNN [163]	20	2021	Amazon	Rmse	✓	3	
ZARM [179]	7	2021	Amazon	Mse	✓	3	
APSE [165]	5	2021	Amazon	Rmse		4	
MPRS [46]	19	2021	Amazon	Mse		2	
UARM [137]	7	2021	Amazon, Yelp	Rmse		2	
MRCP [95]	6	2021	Amazon, Yelp	Mse		4	
NRCMA [107]	5	2021	Amazon	Mse		2	
MDRR [41]	8	2021	Amazon, Yelp	Rmse, Mse, Precision		3	
CARE [134]	5	2020	Amazon, Trip Advisor	Dcg, Ndcg		2	
HTI [154]	4	2020	Amazon	Rmse, Mae		7	
NRCA [96]	12	2020	Amazon	Mse		7	
DANN [176]	29	2020	Amazon	F1, Ndcg	✓	3	
CATN [187]	89	2020	Amazon	Mse	✓	2	
SSG [56]	26	2020	Amazon, Yelp	Mae	✓	5	
JDRM [76]	10	2020	Yelp	Rmse		5	
ARTAN [90]	1	2020	Amazon, Yelp	Mse		6	
AFRAM [88]	24	2020	Amazon, US Online Relailer	Mse, Mae		5	
TextOG [173]	6	2020	Amazon	Mse		3	
HRDR [97]	76	2020	Amazon, Yelp	Rmse	✓	6	
AHN [49]	35	2020	Amazon	Mse		4	
AGTR [175]	9	2020	Amazon, Yelp	Mse, Mae	✓	9	
NRPA [98]	44	2019	Amazon, Yelp	Mse		4	

(Continued)

Table A1. Continued

Name	Citations	Year	Datasets	Metrics	Code	Baselines	Parameters
CARL [159]	167	2019	Amazon	Mse	✓	6	
MLFI [53]	3	2019	Amazon, Yelp	Mse	✓	5	
DAML [94]	128	2019	Amazon	Mse	✓	6	
CARP [85]	97	2019	Amazon, Yelp	Mse	✓	8	✓
RGM [156]	35	2019	Amazon, Yelp	Rmse	✓	4	
CDAE [164]	15	2019	Amazon	Mse	✓	3	
MDR [127]	12	2019	Amazon, Yelp	Mse		2	
SRCMF [151]	22	2019	Yelp	Rmse, Mae	✓	2	
HAUP [166]	56	2019	Amazon, Yelp	Rmse, Ndcg		5	
RSBM [15]	11	2019	Amazon, Yelp	Mse		3	
NCEM [54]	8	2019	Amazon, Yelp	Mse		5	
CCANN [87]	6	2019	Trip Advisor	Rmse		2	
SDRA [42]	48	219	Amazon, Yelp	Mse		4	
RC-DFM [55]	102	2019	Amazon,	Rmse, Mae		0	
DEAML [57]	104	2019	Amazon, Yelp	Rmse		4	
CCoNN [12]	6	2018	Amazon	Rmse		1	
SentiRec [67]	38	2018	Amazon	Mse		2	✓
A3NCF [28]	197	2018	Amazon, Yelp	Rmse	✓	4	
PARL [158]	29	2018	Amazon	Mse		4	
JST [92]	19	2018	Apple	Recall		2	✓
SDNet [27]	39	2018	Amazon	Mse		3	
MPCN [140]	282	2018	Amazon, Yelp	Mse	✓	3	
ANR [30]	158	2018	Amazon, Yelp	Mse	✓	3	✓
NeuO [168]	15	2018	Amazon, Yelp, Taobao	Hr, Mse		1	
ALFM [29]	268	2018	Amazon, Yelp	Rmse	✓	5	✓
NARRE [21]	469	2018	Amazon, Yelp	Rmse	✓	2	
Chaurasiya et. al. [20]	10	2018	Amazon	Rmse, Recall		0	✓
NGMM [45]	19	2018	Amazon	Mse		3	
TARMF [104]	167	2018	Yelp, Amazon	Mse		4	✓
STAR [14]	90	2017	Crawler	Mae, Rmse , Ndcg		2	✓
RB-JTF [135]	43	2017	Amazon	Rmse, Mae		2	
SULM [9]	119	2017	Yelp	Precision, Auc	✓	2	✓
D-ATTN [126]	441	2017	Amazon, Yelp	Mse	✓	2	
ECDR [19]	18	2017	Movies	Precision, Recall, F1		0	
LDA-based [93]	17	2017	Airbnb	Recall		2	✓
UPCF [109]	49	2017	Dianping	Rmse, Precision, Recall , F1		0	✓
AGSG [59]	21	2017	Yelp	Precision, Recall		1	✓
DeepCoNN [189]	942	2017	Amazon, Yelp, Ratebeer	Mse	✓	3	
RCMiner [118]	11	2017	Yelp	Rmse		2	✓
PSR [120]	18	2017	Movieles, Netflix, Eachmovie	Rmse, Mae, Mrr, Ndcg			✓
Transnets [17]	261	2017	Amazon, Yelp	Mse	✓	1	
Multi-Criteria [112]	84	2017	Amazon, Yelp, Trip Advisor	Accuracy		0	✓
RBLT [138]	152	2016	Amazon	Mse		3	✓
ABSA [3]	65	2016	Amazon	Accurac	✓	0	
ITLFM [183]	51	2016	Amazon, Beeradvocate	Mse		6	✓

(Continued)

Table A1. Continued

Name	Citations	Year	Datasets	Metrics	Code	Baselines	Parameters
RPS [83]	119	2016	Yelp	Rmse, Mae		2	✓
ConvMF [78]	795	2016	Amazon, Movielens	Rmse	✓	3	
LRPPM [26]	84	2016	Amazon, Yelp	F1, Ndcg	✓	1	✓
RBSA [5]	27	2016	Cornell	Accuracy		0	
CMLE [184]	87	2016	Crawker	Mse , Mae		4	✓
RHCF [71]	54	2015	Buzzillions	Mae, Precision, Recall, F1		0	✓
LBSN [61]	12	2015	Yelp	Precision, Recall		0	✓
Chen et. al. [24]	75	2015	Yelp, Trip Advisor	Hr, Mrr		2	✓
Sharma et. al. [128]	25	2015	Crawler	Accuracy, Recall, Precision, Error, F1		0	
UGC [167]	12	2015	Yelp, Movielens	Rmse		2	✓
GM [89]	19	2015	Yelp, Trip Advisor	Mse, Rmse		1	✓
CHI [23]	64	2014	Yelp, Trip Advisor	Mrr		4	✓

B Algorithm's Description

- *DeepCoNN (2017)* [189]: leverages CNNs along with maximum pooling to diminish dimensionality and capture pertinent features from comments. It concatenates user and item representations, subsequently employing multiple factorizations to further reduce dimensionality until reaching an estimated rating value.
- *ConvMF (2016)* [78]: employs CNNs to extract pertinent information from comments, thereby creating representations for items. For users, collaborative filtering decomposes the rating matrix into several dimensions, generating representations. Subsequent pooling operations are conducted until an estimated rating value is derived.
- *NARRE (2018)* [21]: also employs CNNs to diminish text dimensionality and utilizes attention mechanisms to discern key aspects constituting user/item representations. It classifies users providing good comments and items that receive those comments, thereby generating reputations for explanatory purposes. Ultimately, neural factorization machines conduct successive dimensionality reductions until the estimated rating value is reached.
- *D-ATTN (2017)* [126]: utilizes two distinct neural networks to generate representations. Both networks employ convolutional layers for dimensionality reduction and feature extraction, differing in their application of attention mechanisms. The first network identifies globally significant aspects of the text, while the second focuses on local aspects. The resulting representations are concatenated and processed by fully connected layers to produce the final user/item representation. Finally, these representations undergo multiplication and addition operations to obtain the estimated rating value.
- *MPCN (2018)* [140]: utilizes a multi-point co-attention architecture, applying multiple attention filters to the text to extract the most relevant comments. Subsequently, word-level attention mechanisms isolate the most significant words within these relevant comments. Finally, the user and item representations are combined, undergoing dimensionality reductions to estimate the rating.
- *ALFM (2018)* [29]: employs an **aspect-aware topic model** (**ATM**) to extract primary topics from all comments, followed by attention mechanisms to isolate key topics within the corpus. Using Jensen–Shannon divergence, it classifies each comment into its relevant topic and constructs user and item representations accordingly. Finally, it conducts multiplication and summation operations between these representations to estimate the rating value.

— *TransNet (2017)* [17]: utilizes two separate convolutional networks: one for extracting document and the other for extracting comment-related characteristics pertinent to the user-item interaction, for which the estimated rating is sought. It computes the discrepancy between the representations derived from the document and the comment to guide the network in estimating the user's potential comment on the item. Once the interaction comment is estimated, it is linked with the user representation and concatenated with the item representation. These concatenated representations then undergo successive dimensionality reductions until a single value is obtained, corresponding to the rating prediction.
— *A3NCF (2018)* [28]: employs TM to extract the primary aspects of the dataset globally. It then diminishes the dimensionality of these topics using convolutional layers, representing users and items based on the comments and feedback they have received, respectively. Subsequently, three consecutive attention mechanisms are applied to discern the principal latent factors within the representations. Finally, fully connected layers are employed to estimate the value of the rating.
— *TARMF (2018)* [104]: utilizes recurrent layers to capture contextual factors among comments, acknowledging that the representation of each word depends on preceding words. Subsequently, attention mechanisms are applied to identify the principal contextual factors for user/item representations. Finally, matrix decomposition is employed on the rating matrix to derive low-dimensional representations of users and items. These representations are then used to compute approximations, adjusting network parameters to obtain the rating value.
— *ANR (2018)* [30]: extracts primary aspects of the text using convolutional layers and attention mechanisms. Subsequently, it computes an affinity matrix between the terms of user and item representations. Sums and projections are then applied to the concatenated representations to estimate the rating.
— *CARL (2019)* [159]: This method employs convolutional layers to diminish the dimensionality of documents and attention mechanisms to extract primary textual aspects. Subsequently, it conducts successive dimensionality reductions on user/item representations individually. The distance between these reduced representations is then calculated to derive the rating value.
— *DAML (2019)* [94]: introduces a learning approach utilizing convolutional layers and attention mechanisms to diminish and extract features from comments. It employs a mutual attention strategy, utilizing an affinity matrix to assess the similarity between all aspects of the user's representation and the item's. Successive projections are then executed to estimate the rating value.
— *CARP (2019)* [85]: employs convolutional layers to reduce dimensionality and extract document features. It introduces a novel mechanism called a capsule network, where different text segments are grouped and classified using attention mechanisms. The network learns to categorize various text segments as positive or negative. Finally, the predicted rating is calculated by estimating the affinity between the user and the item through the disparity between the capsules identified as positive for the user and for the item.
— *CARM (2021)* [86]: This approach employs convolutional layers to diminish the dimensionality of texts and computes the probability distribution among the observed ratings. It then calculates confidence matrices, projecting text representations through dimensionality reduction into the latent space of the ratings, thus enabling the estimation of ratings.
— *HRDR (2020)* [97]: employs convolutions to diminish the dimensionality of texts and projections onto the rating matrixes for further dimensionality reduction. Subsequently, the two latent representations are merged, and attention mechanisms are applied to extract primary aspects. Finally, user representations are combined, and a latent factor model algorithm is applied to predict the estimated rating value.

— *MAN (2023)* [171]: employs two distinct neural networks for recommendation. The first one utilizes convolutional layers to reduce dimensionality and extract features from the text. It incorporates attention mechanisms to identify primary aspects at both word and sentence levels. The second network is similar but without initial dimensionality reduction. Finally, the representations from both networks are aggregated, and factorization machines are employed to conduct successive dimensionality reductions and derive the final rating.

C Comparison between Neu-Review-Rec and iRev

As mentioned in Section 4, we found just one benchmarking framework for RARs, the Neu-Review-Rec [98]. In fact, it has a similar structure to ours. In this sense, we compare the proposed *iRev* framework and Neu-Review-Rec based on five requirements:

(1) Datasets: While Neu-Review-Rec upports only one dataset, *iRev* natively supports six datasets, with the flexibility to easily add new ones, including a tutorial on how to do so.
(2) Recency: Neu-Review-Rec's most recent method is from 2019, while in *iRev* is from 2023.
(3) Coverage: While Neu-Review-Rec implements only five state-of-the-art algorithms (DeepCoNN, DAML, MPCN, DATTN, and NARRE), *iRev* implements 15 algorithms (including those mentioned above), in addition to two adaptations for data input, being BERT embeddings (zero-shot and fine-tuning) for each model, totaling a significantly more extensive range of input/model options and variations. Furthermore, regarding metrics, New-Review-Rec has only three evaluations (MSE, RMSE, and MAE), while *iRev* implements the three previously mentioned plus five additional metrics (Precision, recall, F1, diversity, and novelty), allowing a broader evaluation in terms of algorithm aspects.
(4) Flexibility: Neu-Review-Rec has a rigid implementation structure and requires complex adjustments, such as routine improvements that use inefficient loops instead of linear algebra, compromising flexibility and performance. On the other hand, *iRev* follows a similar structure but was designed to facilitate the instantiation of new experiments and integration of components without significant changes to the code. Therefore, *iRev* can be considered a less rigid framework that allows users to modify without major changes.
(5) Extensiblity: In line with the previous point, finally, extensibility: While Neu-Review-Rec can be considered extensible, there are still code limitations since more advanced customizations require in-depth modifications. On the other hand, *iRev* is fully extensible, focusing on modularity and component reuse. It was created to allow easy extension with new algorithms, metrics, and datasets without compromising simplicity of use, counting on templates for ease of extension.

In summary, *iRev* is superior to Neu-Review-Rec regarding the number of algorithms and metrics, ease of instantiation, and extensibility, providing a more versatile and powerful platform for experiments and customizations.

D Detailing of Architectures

Table A2. Learning Architectures: Characteristics, Advantages, Disadvantages, and Application

Method	Characteristics	Advantages
Matrix Decomposition	incorporate users' comments through joint factorization techniques for adding textual representations as additional features.	Effective in handling sparse data.
Clustering	Group users or items based on similarity based on characteristics derived from comments.	Identify patterns of behavior or preferences in groups of users.
Standard Graph	Consider interactions between users, items, and comments as a graph, where nodes represent users, items, or comment keywords, and edges represent interactions or similarities.	Capture complex interactions between users, items, and the relationships between comment words.
Convolution	Apply convolutions to capture local patterns in comment texts, such as key phrases and n-grams, extracting useful textual representations for recommendation.	Effective in capturing local patterns and contextual semantics of comments.
Attention	Use attention mechanisms that allow the model to focus on specific parts of the comment text that are most relevant for the recommendation.	Effective in modeling the complex semantics and general context of comments. Capture long-range dependencies in the text, considering the contextual relevance of each word or phrase.
Recurrent	Model the temporal sequence and structure of longer reviews, considering word order.	Capture temporal and long-range dependencies in the text, which is useful for understanding the progression of reviews in long reviews.
Neural Graph	Use the graph structure to propagate information between nodes, extracting rich embeddings that capture both the connections between users and items and the textual content of reviews.	Efficiently capture the complex structure of interactions and relationships between users and items, incorporating both explicit and implicit feedback. Adaptable to different types of data (textual, relational, social) within the same structure.
	Applications	**Disadvantages**
Matrix Decomposition	Streaming or online scenarios where recommendation needs to be fast and massive.	Do not capture non-linear interactions or complex contexts of textual comments well.
Clustering	Scenarios where group behavior can define users' preferences well, such as social networks.	Challenge in defining similarity metrics appropriate for textual data and difficulty in dealing with high-dimensional data.
Standard Graph	Social and collaborative recommendation networks, where interactions between users, items, and textual comments can be captured.	More complex implementation and higher computational cost due to the dimensionality of textual data and difficult to interpret due to the highly connected and interactive nature of graphs.
Convolution	Short text. Recommender systems that benefit from local patterns in short comments.	Limited in capturing global dependencies in text sequences and broader contexts.
Attention	Systems that need to interpret comments with a high level of detail and with large contextual diversity.	More complex implementation and greater need for hyperparameter optimization.
Recurrent	Media platforms that need to understand the narrative or progression of ideas in user comments.	High computational cost and it might face vanishing gradient issues.
Neural Graph	Recommendation platforms based on complex user and item interactions with detailed reviews and multiple layers of interaction representation.	Complex implementation and high computational cost, especially for large graphs. Requires a large amount of data to generate high-quality embeddings.

E Comparing RARs to Traditional Recommender Systems

We performed a comparative analysis among the best review-based Recommender Systems (RARS) (Table 11) and traditional recommendation approaches that do not use reviews in the inference process. The selection of traditional algorithms was based on studies such as [8, 177], allowing us to compare three different classes of algorithms, totaling five traditional RSs: two widely recognized matrix factorization methods (**Non-negative Matrix Factorization** (**NMF**) and **Singular Value Decomposition** (**SVD**)), two explicit collaborative filtering and community analysis algorithms (**User-based K-Nearest Neighbors** (**UserKNN**) and **Item-based K-Nearest Neighbors** (**ItemKNN**)), and a **neural collaborative filtering** (**NCF**) recommendation model. We detail them below.

(1) **NMF** [108] is an MF technique that decomposes the user-item interaction matrix into two lower-dimensional matrices, keeping only non-negative values.
(2) **SVD** [190] is a factorization technique that decomposes the interaction matrix into three matrices that capture complex relationships between users and items.
(3) **UserKNN** [74] relies on similarity between users, recommending items that have been highly rated by users with similar preference profiles.
(4) In **ItemKNN** [73], recommendations are generated based on the similarity between items. Instead of focusing on relationships between users, this algorithm examines items that are frequently rated similarly
(5) **NCF** [66] is a neural network-based approach that uses latent representations of users and items to generate recommendations. Unlike traditional factorization methods, NCF can capture non-linear and complex relationships between users and items.

We evaluated the algorithms on all datasets used in the main experimental analysis of this article. To balance the systems across different dimensions and provide a comprehensive analysis of the usage contexts of each approach, we selected two metrics from each category used in the main experiment. The chosen metrics were: two error metrics (MSE, MAE), two ranking metrics (Precision@10, Recall@10), and two diversity metrics (Diversity, Novelty). Table A3 presents the results obtained for all algorithms, including the Wilcoxon statistical test, to identify statistically significant gains and ties.

To start with, we notice that RARS have demonstrated superior performance regarding accuracy, reinforcing the idea that user comments capture their preferences and consumption patterns in a rich and implicit way. This RARS advantage is further observed in the error metrics in most cases. The success of RARS is not solely due to the use of comments but also to their ability, based on neural architectures, to identify patterns, as evidenced by the good results of NCF in these same metrics. This validation instills trust in the results produced by RARS.

We also observe that traditional algorithms significantly outperform RARS in Recall. One possible explanation is that explicit collaborative filtering algorithms capture user preferences and implicit consumption diversity, as also evidenced by their superior performance in Novelty. RARS algorithms, while highly accurate in understanding consumption patterns, have limitations in recommending additional items that could be relevant, a drawback that could be the subject of further work.

What these experiments reveal is that the classic tradeoff between accuracy and diversity, a well-documented phenomenon in the literature [178], is also present when comparing RARS methods – which excel in accuracy – with traditional methods – able to produce more diverse results. Future work should focus on improving the process of extracting knowledge from reviews and on exploring the potential of combining traditional recommendation methods with review-based strategies.

Table A3. Experimental Results

Amazon											
Dataset	Metric	Review Aware RS					Traditional RS				
		NARRE	CARP	HRDR	CARM	MAN	NMF	SVD	UserKNN	ItemKNN	NCF
Digital Music	MSE	0.321	**0.302**	0.320	0.349	0.321	1.175	1.180	0.950	1.190	1.182
	MAE	0.345	**0.316**	0.336	0.371	0.334	0.816	0.813	0.681	0.710	0.867
	Precision@10	0.497	0.511	0.498	0.451	**0.530**	0.124	0.127	0.020	0.035	0.054
	Recall@10	0.345	0.366	0.333	0.366	0.398	0.749	**0.761**	0.158	0.256	0.411
	Diversity	0.041	0.088	0.051	0.117	0.079	**0.240**	**0.240**	0.206	0.206	0.217
	Novelty	0.043	0.041	0.040	0.044	0.044	0.042	0.051	**0.617**	0.559	0.339
Musical Instruments	MSE	**0.862**	0.884	**0.860**	0.885	0.894	0.905	0.911	1.140	1.295	0.895
	MAE	0.670	0.668	0.653	0.666	0.655	0.692	0.689	**0.645**	0.675	0.697
	Precision@10	0.414	0.434	0.417	0.385	**0.522**	0.065	0.066	0.027	0.028	0.045
	Recall@10	0.256	0.239	0.251	0.245	0.289	0.549	**0.551**	0.260	0.270	0.400
	Diversity	0.071	**0.228**	0.110	0.101	0.090	0.214	0.214	0.202	0.201	0.204
	Novelty	0.075	0.074	0.069	0.069	0.078	0.045	0.047	**0.503**	0.482	0.284
Office Products	MSE	0.922	0.930	**0.863**	0.927	0.886	**0.865**	0.872	0.872	1.077	1.026
	MAE	0.676	0.676	0.703	0.701	0.716	0.723	0.725	**0.620**	0.663	0.809
	Precision@10	0.379	0.373	0.391	0.337	**0.410**	0.104	0.102	0.019	0.033	0.051
	Recall@10	0.248	0.240	0.301	0.255	0.298	**0.659**	**0.659**	0.175	0.284	0.418
	Diversity	0.069	0.162	0.170	**0.270**	0.169	0.230	0.225	0.170	0.171	0.185
	Novelty	0.070	0.071	0.069	0.106	0.061	0.003	0.010	**0.727**	0.606	0.321

Yelp											
Dataset	Metric	Review Aware RS					Traditional RS				
		NARRE	CARP	HRDR	CARM	MAN	NMF	SVD	UserKNN	ItemKNN	NCF
Tucson	MSE	1.786	1.772	**1.698**	1.858	**1.702**	2.195	2.199	2.021	2.562	2.038
	MAE	1.060	1.078	1.057	1.089	**1.001**	1.264	1.264	1.033	1.109	1.097
	Precision@10	**0.365**	0.337	0.337	**0.362**	0.346	0.087	0.087	0.024	0.040	0.047
	Recall@10	0.275	0.265	0.234	0.280	0.190	**0.578**	**0.577**	0.194	0.297	0.353
	Diversity	0.202	0.130	**0.219**	**0.210**	0.122	0.133	0.133	0.134	0.134	0.134
	Novelty	0.100	0.097	0.110	0.133	0.095	0.006	0.014	**0.589**	0.471	0.329
Tampa	MSE	1.631	**1.600**	1.695	1.710	1.629	1.962	1.966	1.839	2.333	1.830
	MAE	0.988	0.977	1.003	1.021	1.001	1.173	1.173	**0.967**	1.049	1.026
	Precision@10	0.372	0.330	0.310	**0.380**	0.295	0.088	0.088	0.022	0.038	0.046
	Recall@10	0.276	0.262	0.274	0.272	0.304	**0.598**	**0.596**	0.180	0.289	0.363
	Diversity	0.133	**0.184**	0.109	0.087	0.111	0.144	0.143	0.130	0.131	0.132
	Novelty	0.093	0.140	0.090	0.105	0.095	0.005	0.013	**0.590**	0.481	0.302
Philladelphia	MSE	1.412	**1.401**	1.451	1.450	1.420	1.721	1.722	1.586	2.069	1.604
	MAE	**0.923**	0.933	0.952	**0.923**	0.976	1.067	1.068	**0.918**	1.009	0.957
	Precision@10	**0.365**	0.335	0.310	**0.363**	0.358	0.100	0.099	0.024	0.040	0.052
	Recall@10	0.267	0.262	0.262	0.268	0.260	**0.639**	**0.633**	0.178	0.296	0.387
	Diversity	0.084	0.222	**0.272**	0.110	0.137	0.145	0.145	0.134	0.134	0.137
	Novelty	0.084	0.083	0.097	0.110	0.091	0.005	0.011	**0.649**	0.491	0.308

F Limitations

We only considered the last 10 years of literature in all the analyses. We did our searches only on Google Scholar, based on the premise that it can index the main digital academic libraries. We did not analyze studies focusing exclusively on using user reviews to explain recommendations. Finally, we conducted an experimental evaluation of the main RSs in the field, initially considering 16 algorithms selected according to their total number of citations, the scope of the experimental evaluations, and their recency. More options can be considered in the future. In any case, all the codes were made available in a benchmark that can be expanded by adding new algorithms.

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